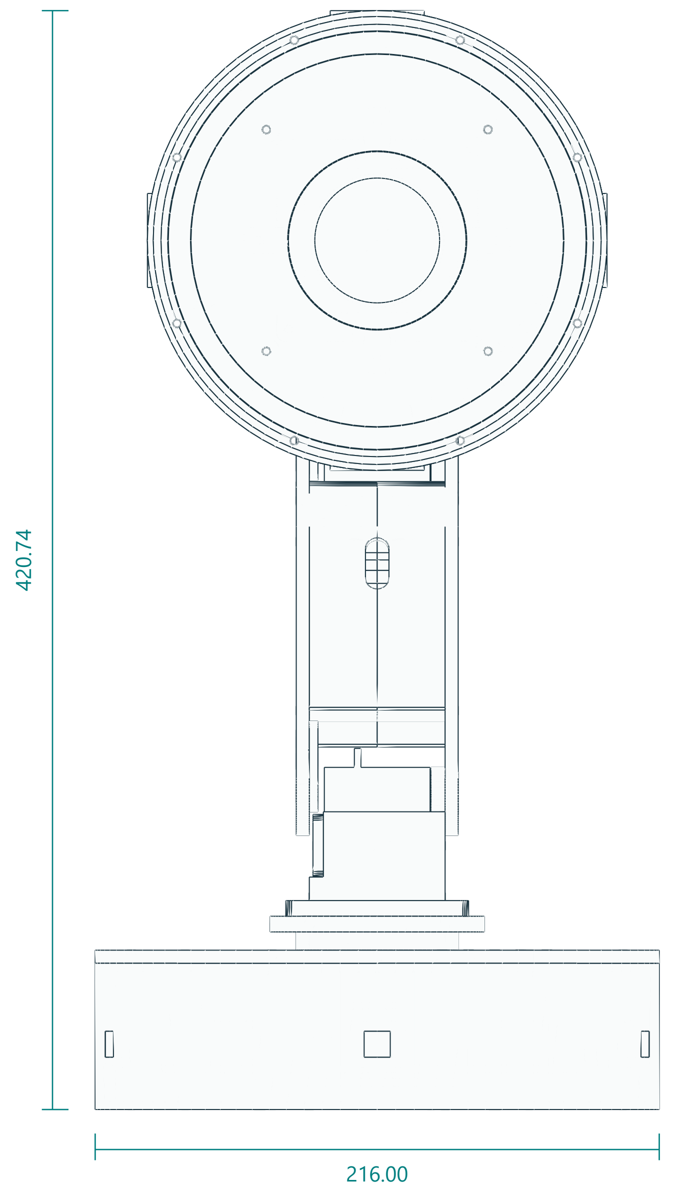
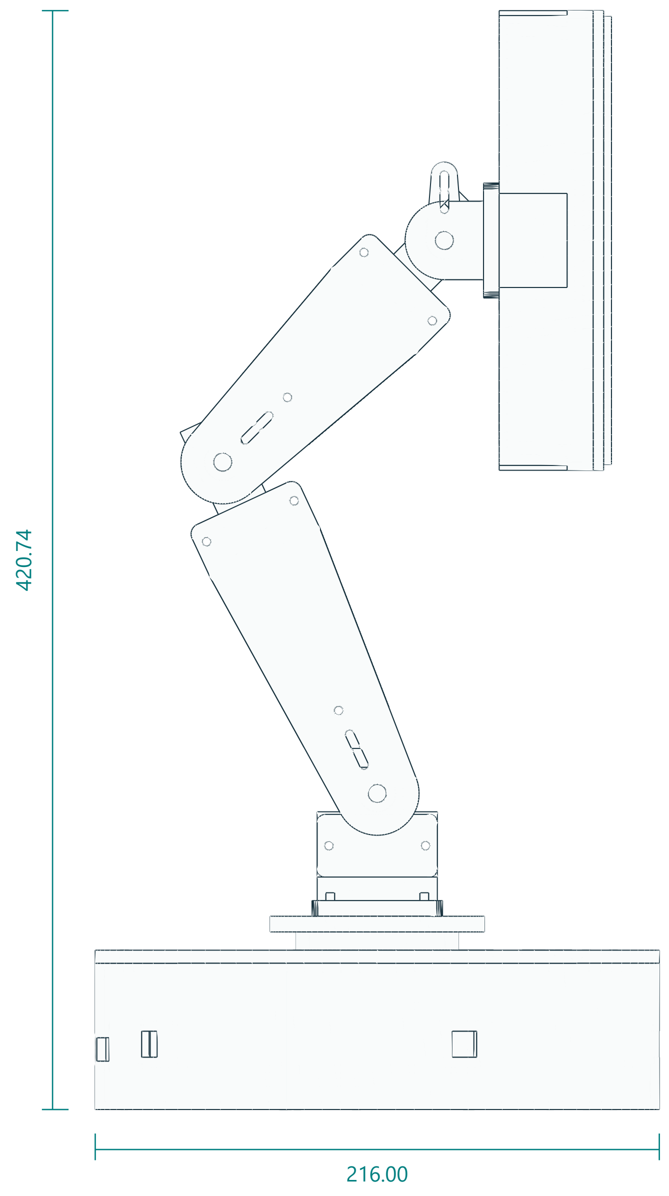
Finished assembly / indexed neutral pose

Assembled geometry from the same STL files supplied for printing. Purchased parts are simplified envelopes.

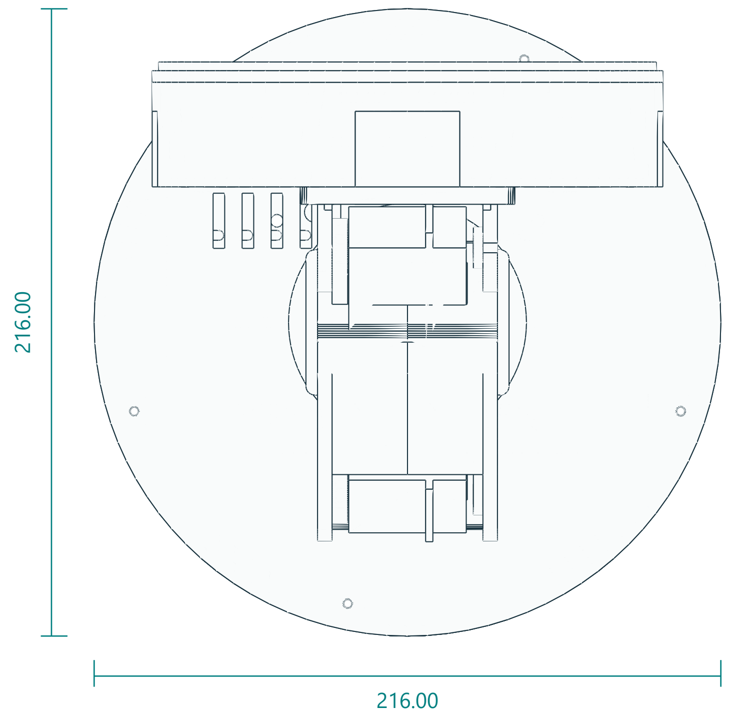
FRONT / LCD FACING VIEW



RIGHT SIDE / NEUTRAL POSE



TOP / FOOTPRINT



Four axes: yaw about Z; shoulder, elbow and wrist about local X. Datum is the base underside at Z=0.
Nominal linkage pitches: shoulder–elbow 140 mm; elbow–wrist 120 mm. The illustrated height is a pose dimension, not a motion envelope.
Neutral: shoulder link 115° above horizontal; forearm 45° above horizontal; face directed toward +Y. Follow indexed calibration in the manual.
Five Adafruit3967 boards mount inside the pedestal wall. Four DS3218MG servos use driven straight horns and opposite625 supports at pitch joints.
Do not infer screw lengths from the illustration. Fasteners and service loops are specified in the assembly chapter.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

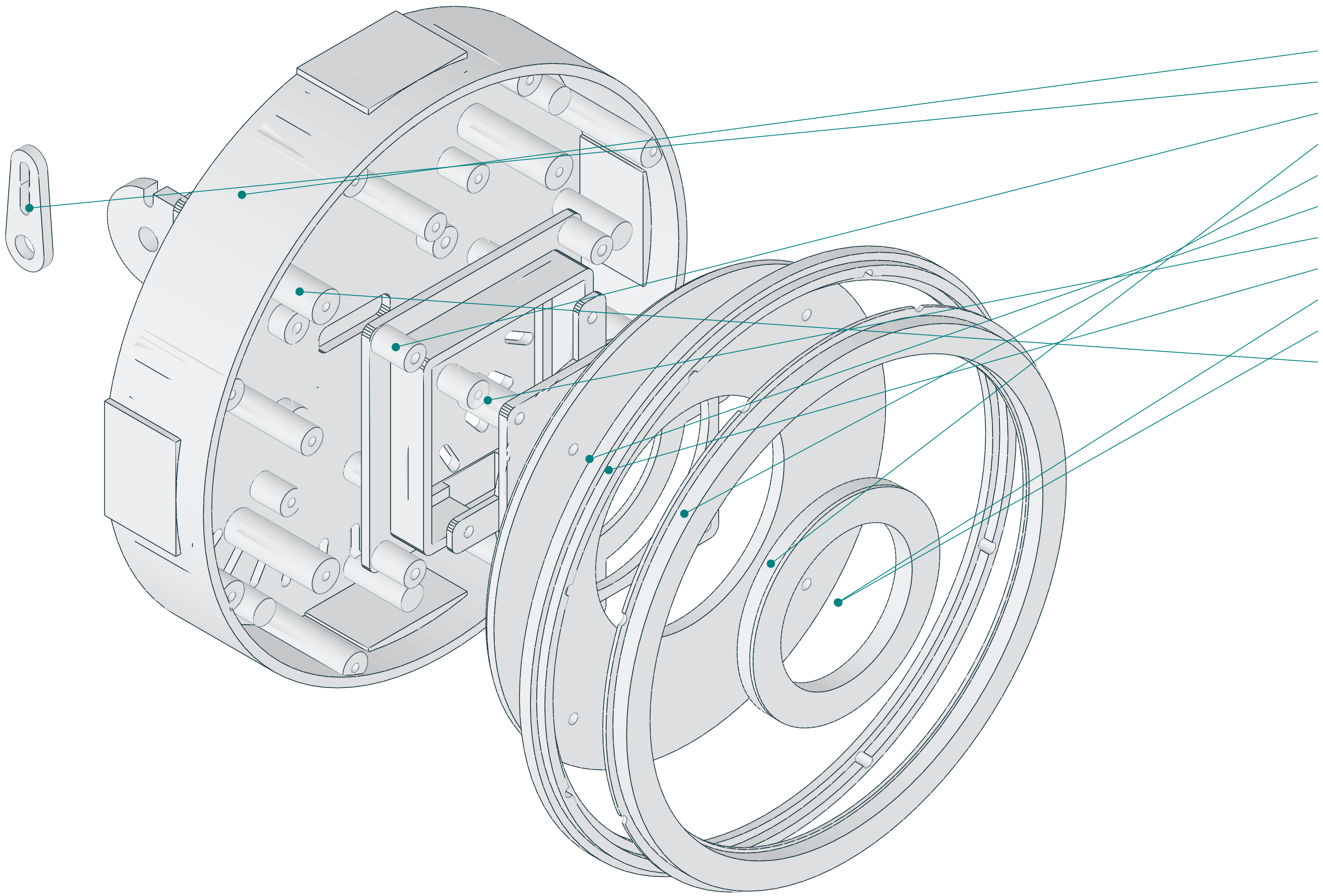
G01

REV C | 07 SEP 2026 | A3

Head / layered assembly study

Exploded positions expose the optical parts; displacements are illustrative and are not insertion paths.

EXPLODED HEAD / MODEL-DERIVED VIEW



LAYER REFERENCES

01 Head tilt fork
02 head driven horn spacer
03 Head Shell
04 Outer Bezel
05 Face Center
06 White Carrier
07 Lcd Cradle
08 Lcd Retainer
09 Outer Diffuser
10 Inner Diffuser
11 Purchased LCD envelope

Central LCD board is behind the inner white ring. Keep all optical components at their documented axial heights.
Use natural PETG for both light diffusers and opaque PETG for the optical baffle. Leave LCD glass clear; all ToF apertures are in the pedestal.
The four outer ring PCB quarters require continuous mechanical support. Solder joints are electrical connections only.
Refer to the head assembly chapter for screw order, spacers, wire relief and board retention. Purchased LEDs are not printable parts.

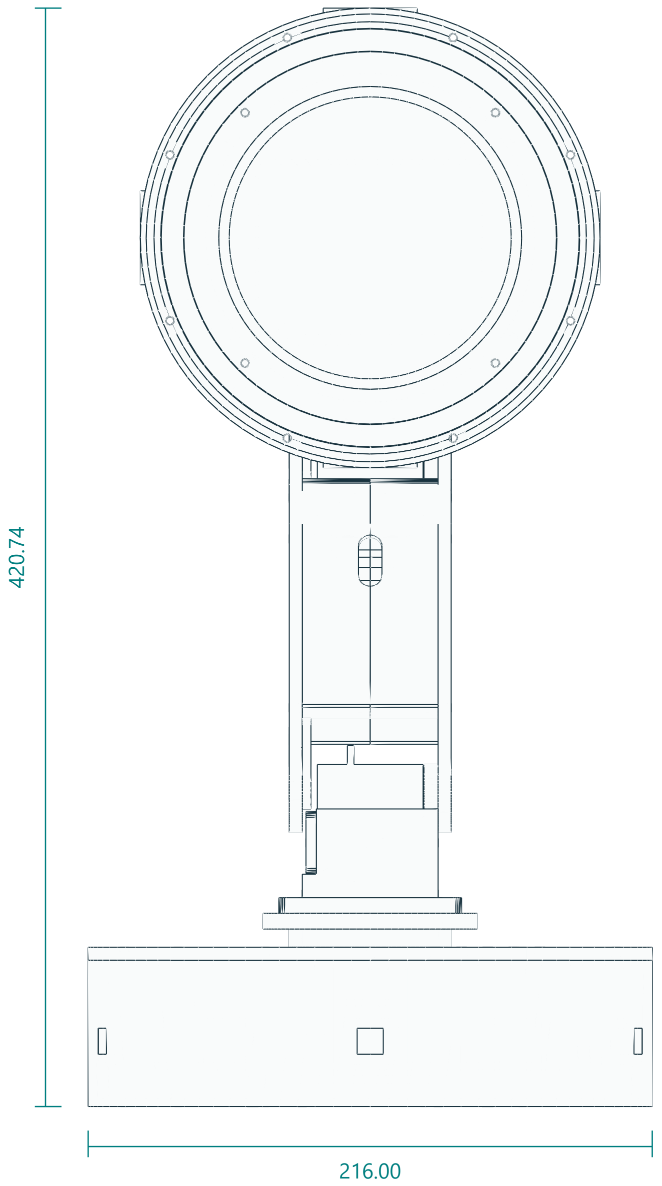
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

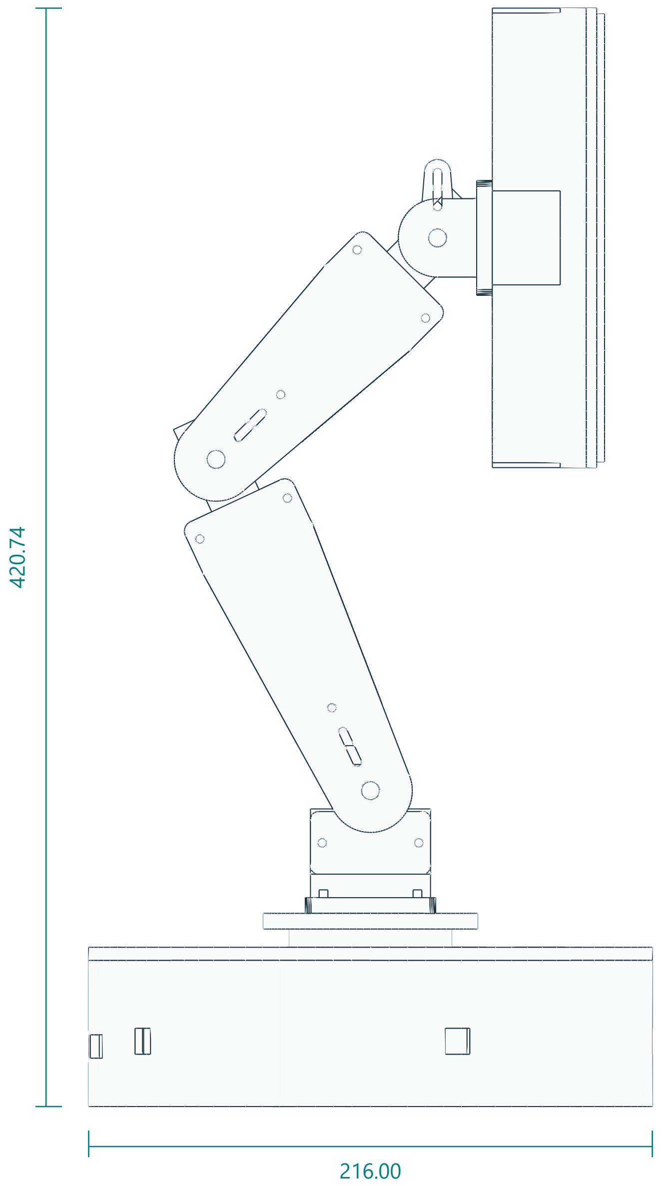
Optional 4-inch DSI head / indexed neutral pose

Same sensor-equipped pedestal, arm links and fork as G01. Head shell, carrier and face ring are the optional parts; the display envelope is measured from the vendor STEP.

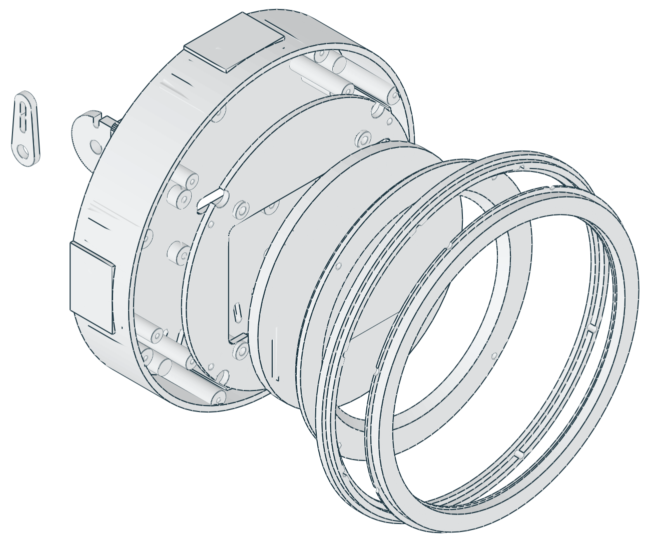
FRONT / LCD FACING VIEW



RIGHT SIDE / NEUTRAL POSE



EXPLODED DSI HEAD



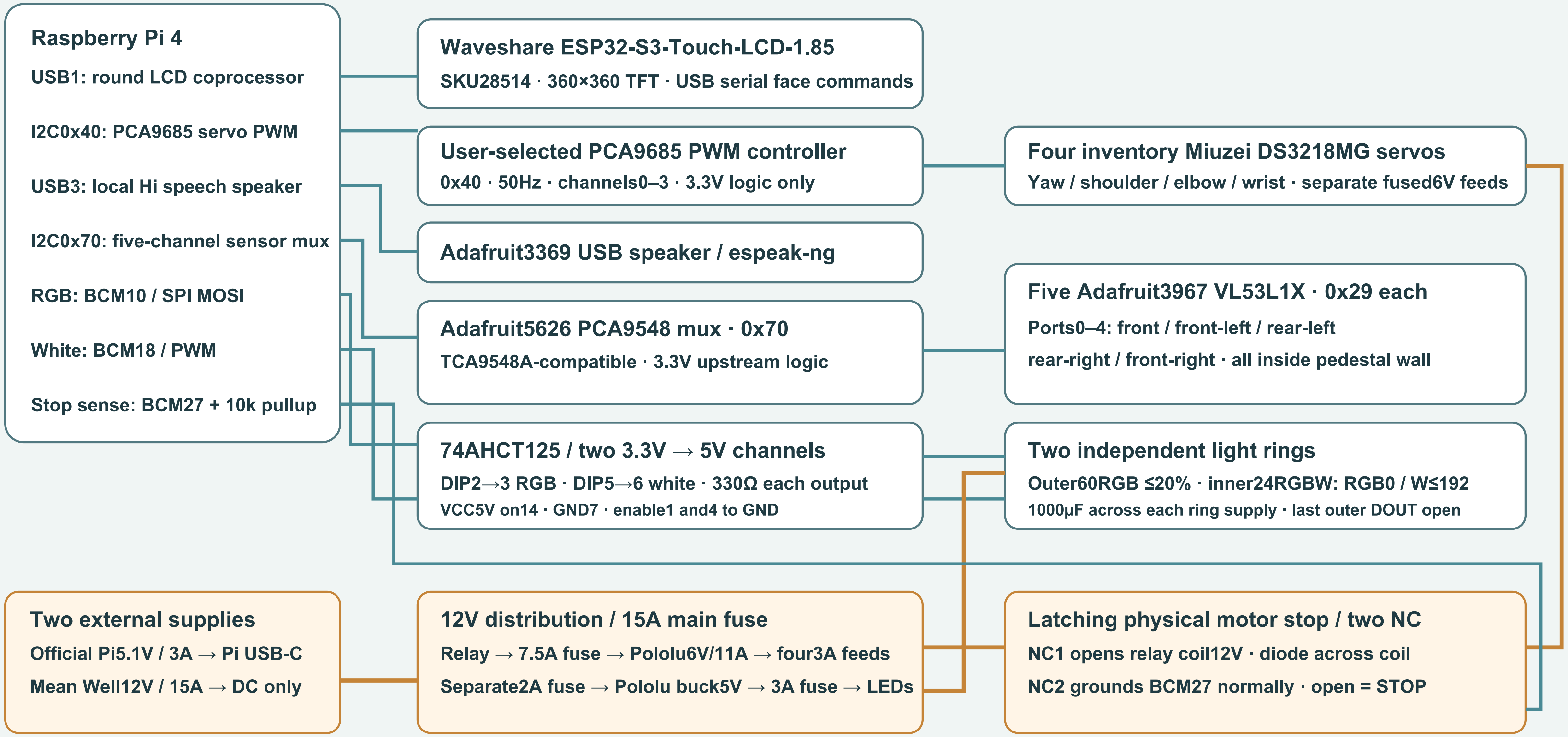
Waveshare 4inch DSI LCD (C): Ø126 case, 6 mm thick, four M4 bosses on a 75 × 75 pattern. Rim front at head Z35.5; carrier plate Z20.5–23.5 with pads to the boss ends at Z25.5.
Face ring opening Ø108 chamfered to Ø116 leaves a 3 mm black rim around the Ø101.5 active area. Outer60-RGB halo, bezel, diffuser and fork are shared with the standard head.
No inner white ring on this head: the halo provides capped white output. Revision C has no head-mounted sensors or brow bracket.
Cables: 15-pin DSI FFC plus 5 V/GND leave through the 26 × 8 slot in the shell floor and fork plate, then run inside the enclosed links through their wall ports.

Electrical connections / system wiring

Follow the point-to-point wiring schedule for connector pins, polarity and conductor ratings.

LUMA / Electrical connections

Revision C · Raspberry Pi 4 master · four PWM servos · five pedestal-wall ranging sensors



Read wiring.csv for exact connections, fuses, DIP pins and cable notes.

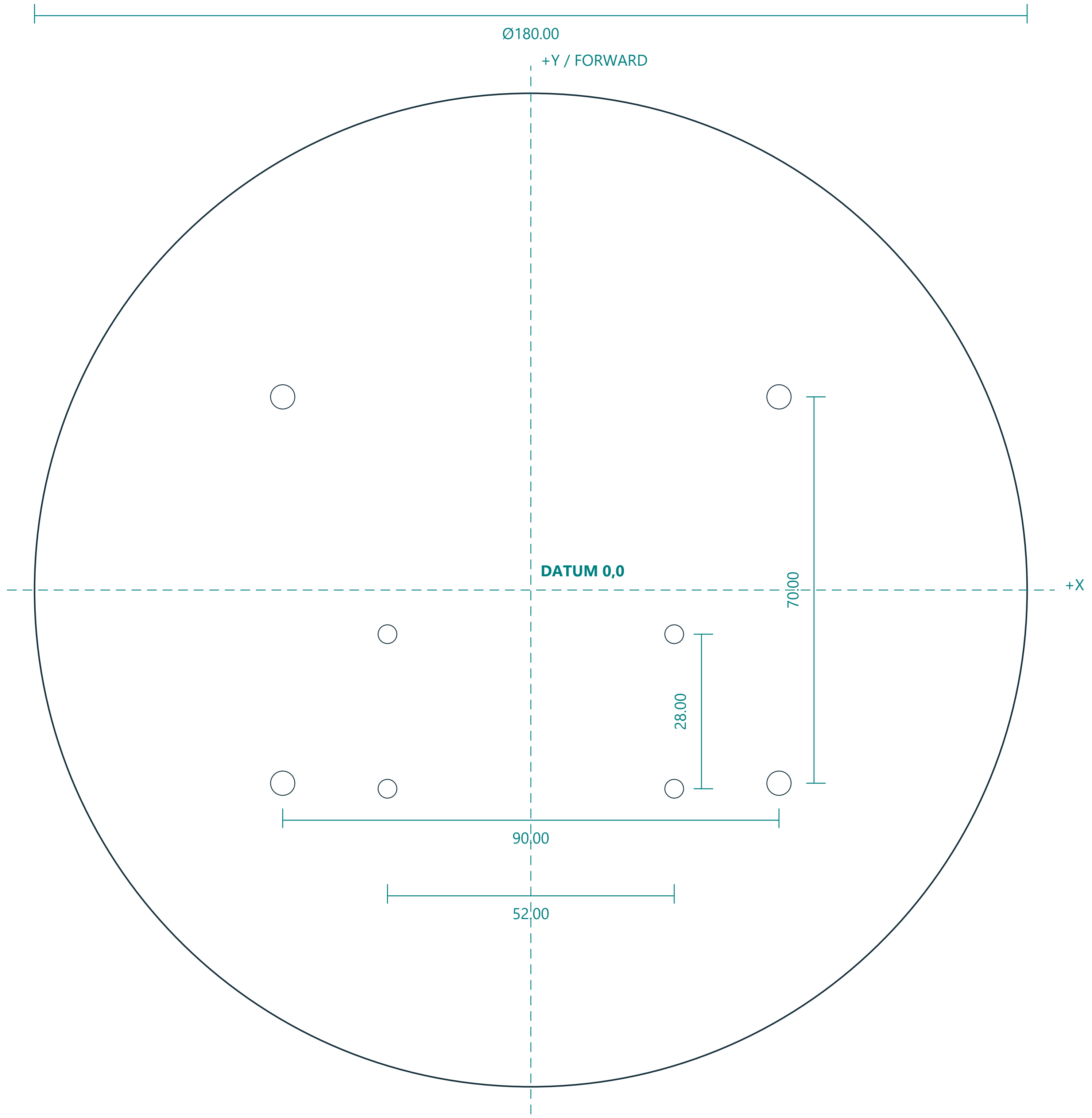
All DC grounds meet at one star point. Never join Pi5V and LED5V. Do not carry servo power through PCA9685 traces.

Motor stop cuts6V power: support the arm. PWM confirms no shaft position; five narrow ToF cones are not a safety perimeter.

Power wires shown in amber; signal routes in blue. Schematic overview: routing lengths and connector positions are not to scale.

Steel ballast plate / fabrication drawing

Purchased/custom-fabricated metal part M01. This sheet is not a printable-plastic substitute.



MATERIAL + INTERFACES

Material: mild steel; finished thickness6.00mm.
Outer diameter180.00mm; deburr all edges and holes.
Datum origin is the disc center. All hole axes normal to plate.
4 × Ø4.40 through: X=±45.00, Y=±35.00.
4 × Ø3.40 through: X=±26.00, Y=-36.00 and-8.00.
Do not add a center hole. Rear yaw bolt pattern points-Y.
Nominal mass before drilling:1.19kg at steel density7.85g/cm^3.
Fit to printed base floor before final coating. Inspect burrs and screw-head clearance.
Suggested workshop positional allowance±0.20mm; confirm actual printed mating pattern before drilling.

SIDE VIEW / THICKNESS

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

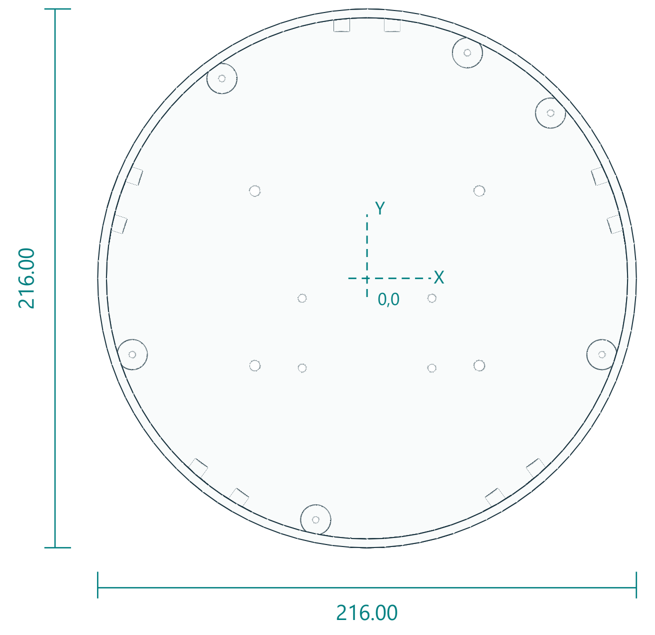
F01

REV C | 07 SEP 2026 | A3

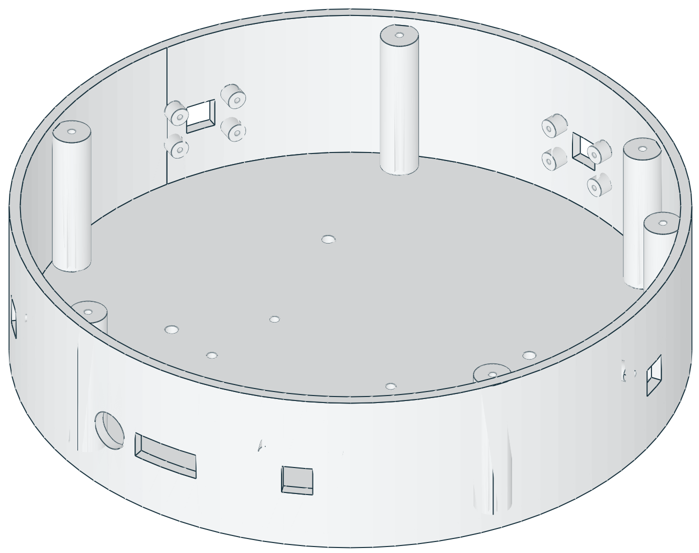
Weighted base enclosure with five internal ToF mounts

base_tub.stl | PRINT QUANTITY 1 | PETG

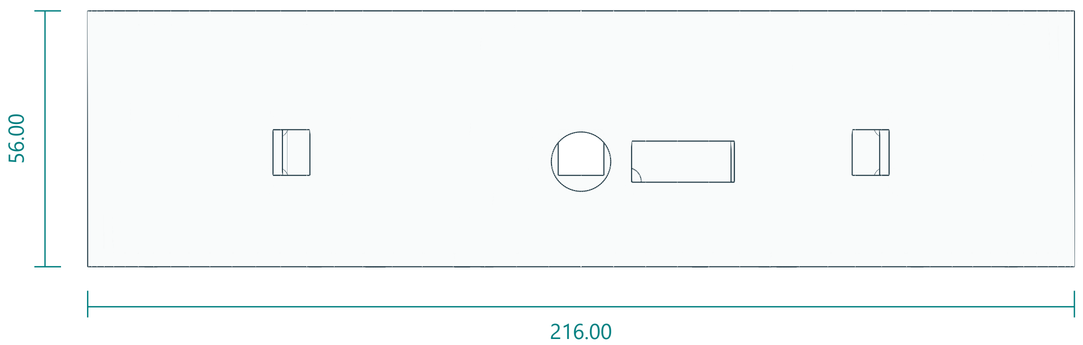
TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



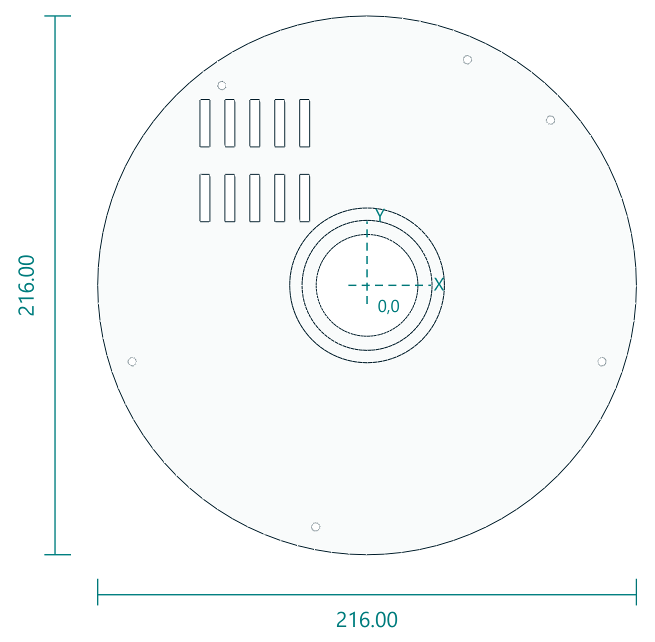
RIGHT / YZ

Interfaces: 216 diameter; 56 high; 3 floor; 3.5 wall; five sensor apertures and 20 internal bosses.
Fasteners / retention: 6 M3x12 lid screws; 4 M4 ballast retainers; 20 M2.5x6 sensor screws.
Print orientation: floor down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 clearance4.4 at (+/-45,+/-35). Yaw bolts3.4 at (+/-26,-36) and (+/-26,-8). Six lid pilots2.8 at radius99, angles42/66/126/198/258/342deg. Five 10x10 wall apertures at18/90/162/234/306deg, Z25. Each board has four2.1 blind pilots on20.32x12.70 at tangent radius99. Rear ports occupy270/282deg.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

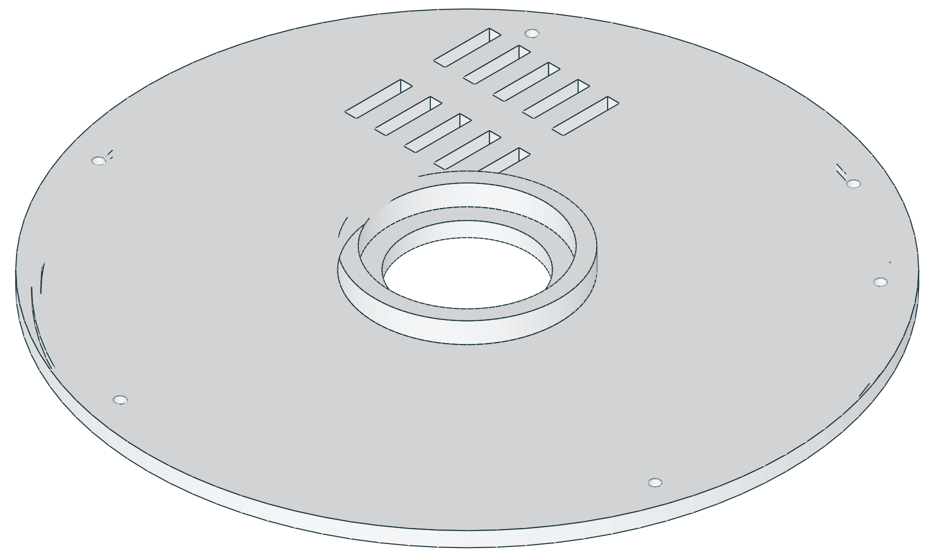
Vented base lid and yaw bearing seat

base_lid.stl | PRINT QUANTITY 1 | PETG

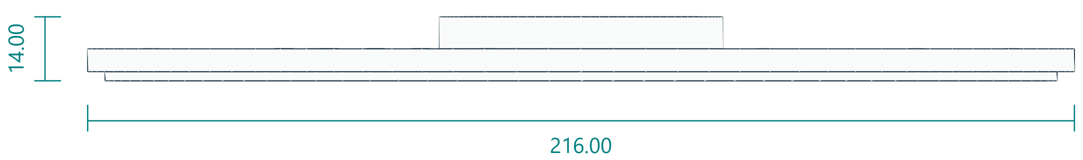
TOP / XY



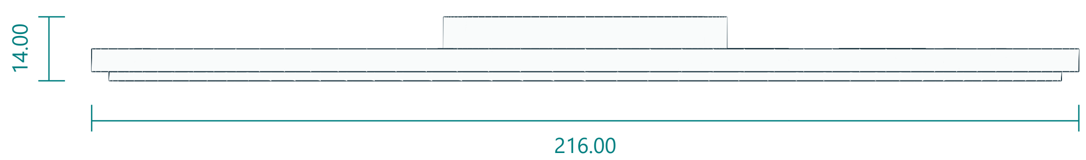
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 216 diameter; bearing pocket52.15 x7; 40.8 cable/hub bore.
Fasteners / retention: 6 M3x12.
Print orientation: flat annulus down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Six3.4 holes radius99 at42/66/126/198/258/342deg. Bearing52.15 bore by7 depth. STL bottom datum is2mm below principal lid underside.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

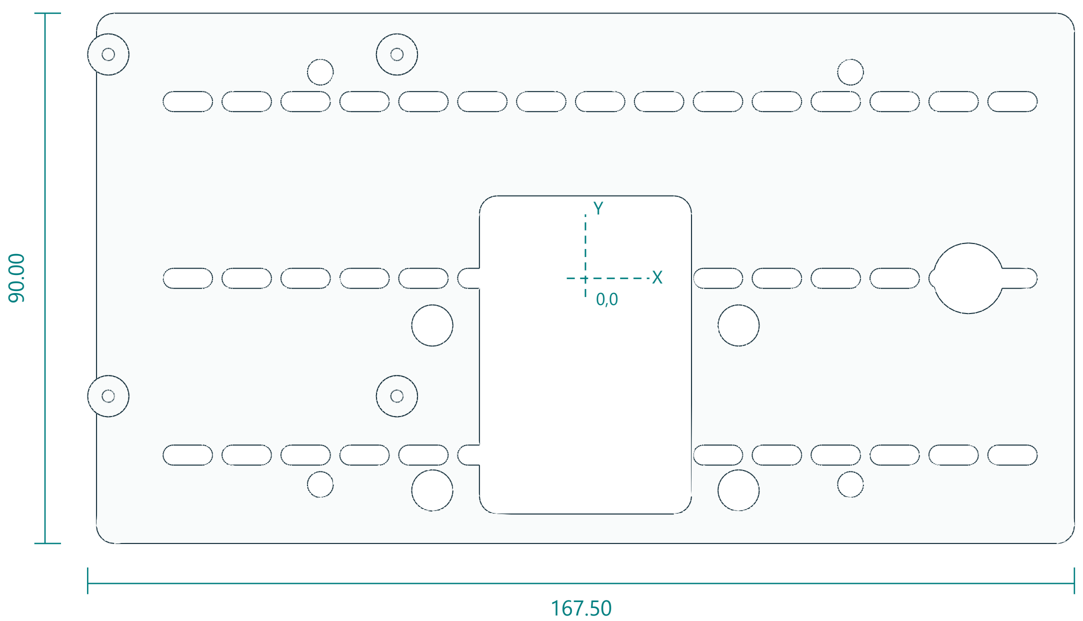
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

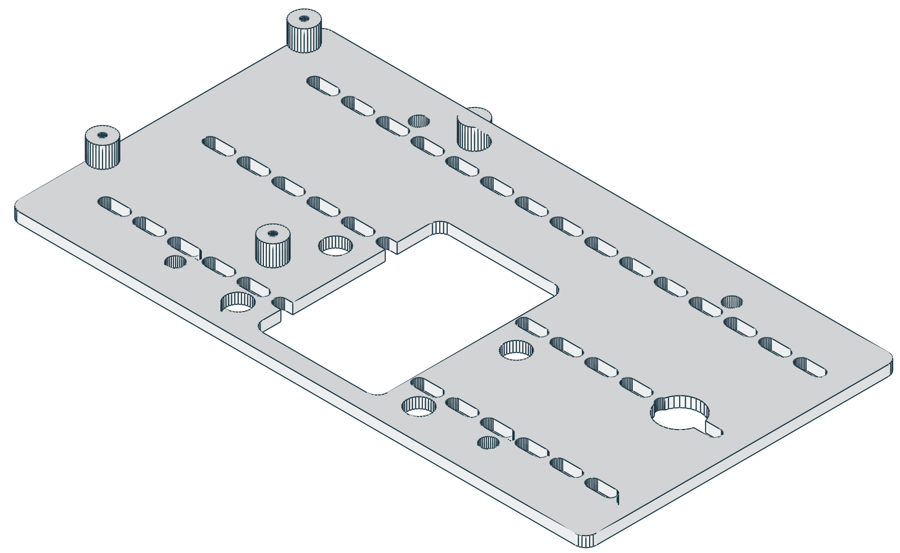
Universal electronics tray with Pi4 standoffs

electronics_tray.stl | PRINT QUANTITY 1 | PETG

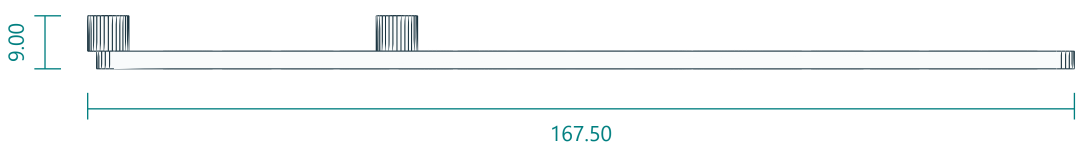
TOP / XY



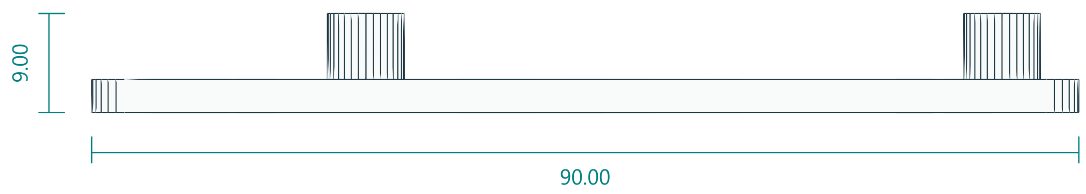
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

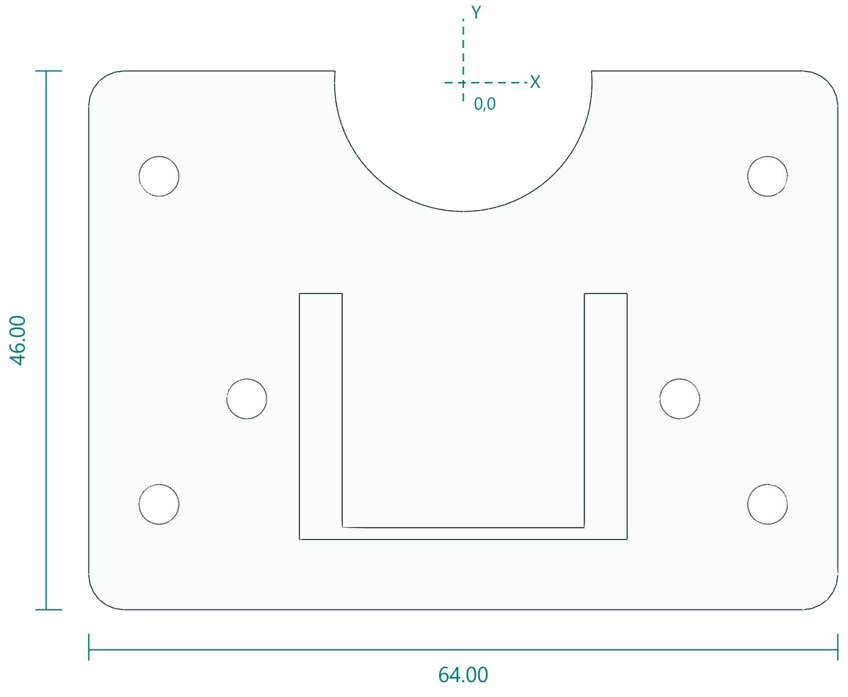


Interfaces: 166 x90; Pi4 hole pitch49 x58; 2.1mm M2.5 tapping pilots.
Fasteners / retention: 4 M2.5x6 Pi; 4 M4x25 with metal spacers.
Print orientation: plate down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 holes (+/-45,+/-35). Pi M2.5 pilots2.1 at X=-81,-32 and Y=-20,38. Central rounded36x54 clearance centered(0,-13). Four7mm tool apertures at yaw bolts.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

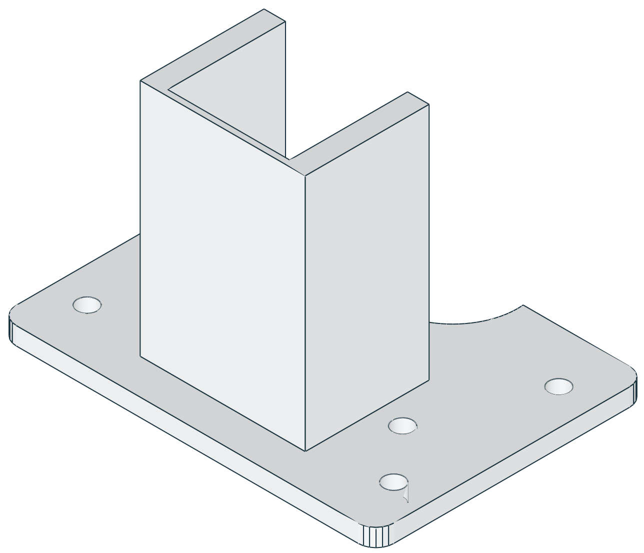
DS3218MG yaw servo rear-body cradle

yaw_mount.stl | PRINT QUANTITY 1 | PETG

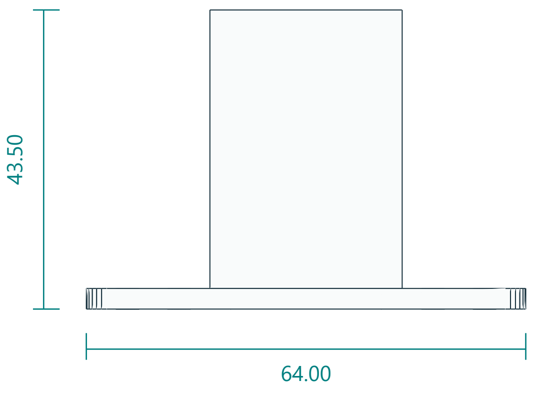
TOP / XY



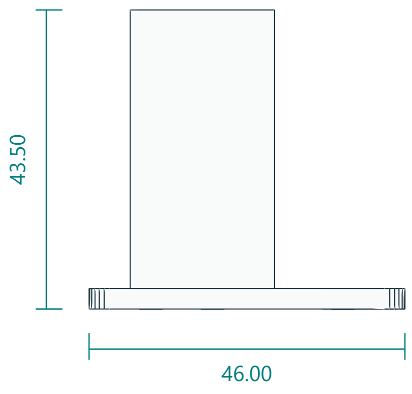
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

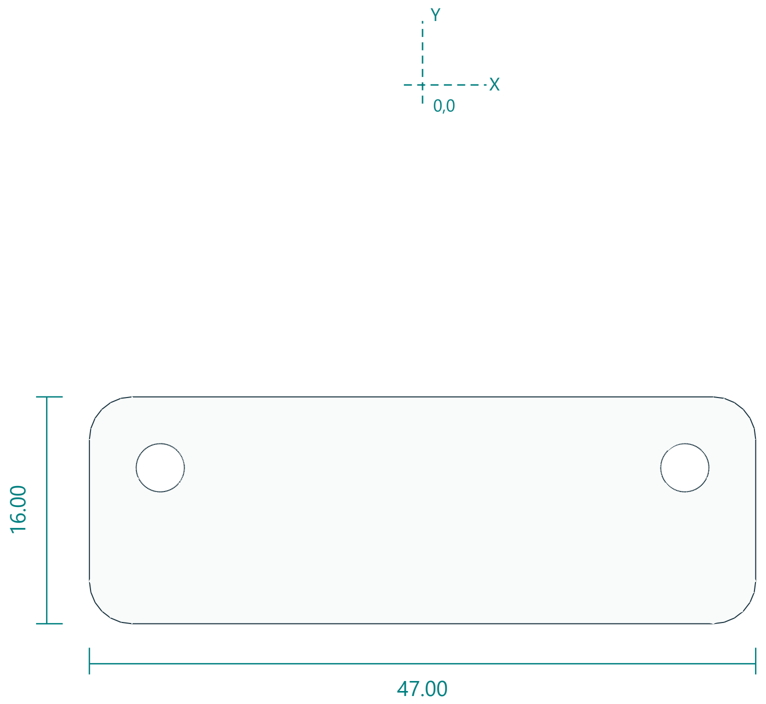


Interfaces: 64 x46; pocket20.7; 40.5-high rear body clamp.
Fasteners / retention: 4 M3x12 mount; 2 M3x50 clamp.
Print orientation: base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 floor holes (+/-26,-36/-8). Keeper bolts at (+/-18.5,-27). DS3218 shaft atXY(0,0), output case plane Z43.5 in part coordinates.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

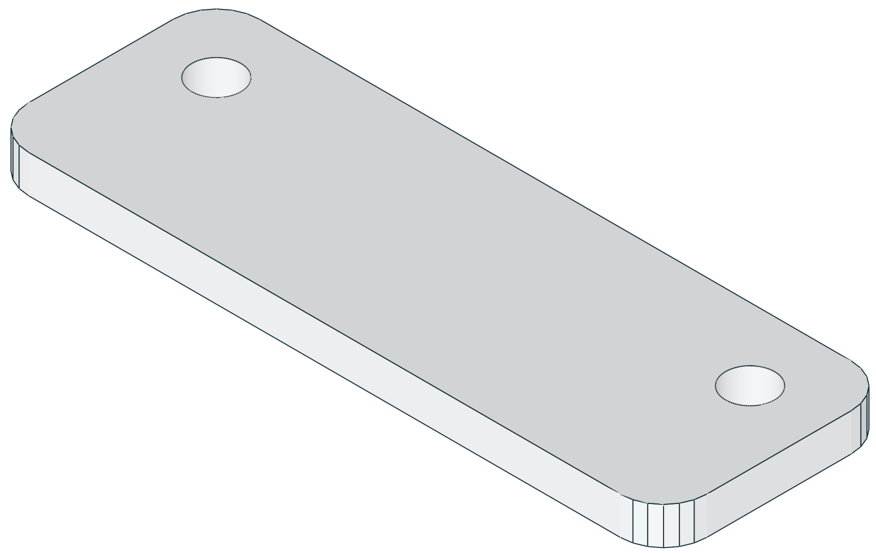
Yaw clamp keeper

yaw_cap.stl | PRINT QUANTITY 1 | PETG

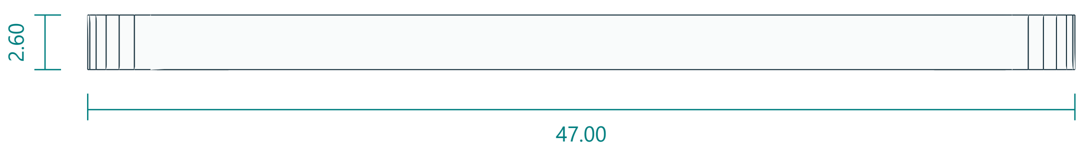
TOP / XY



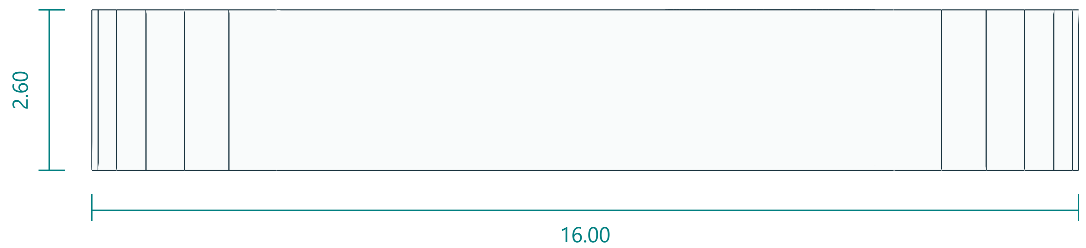
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 47 x16 x2.6; clamp hole pitch37.
Fasteners / retention: 2 M3x50.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,-27). Keeper center(0,-30); thickness2.6.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

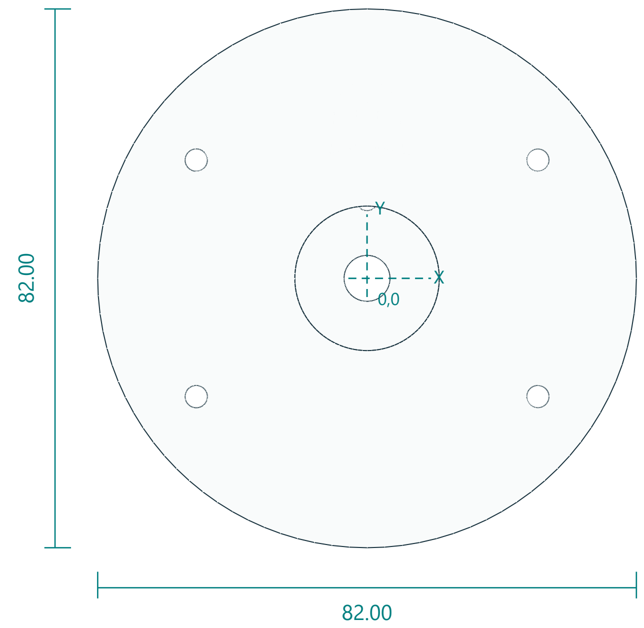
P05

REV C | 07 SEP 2026 | A3

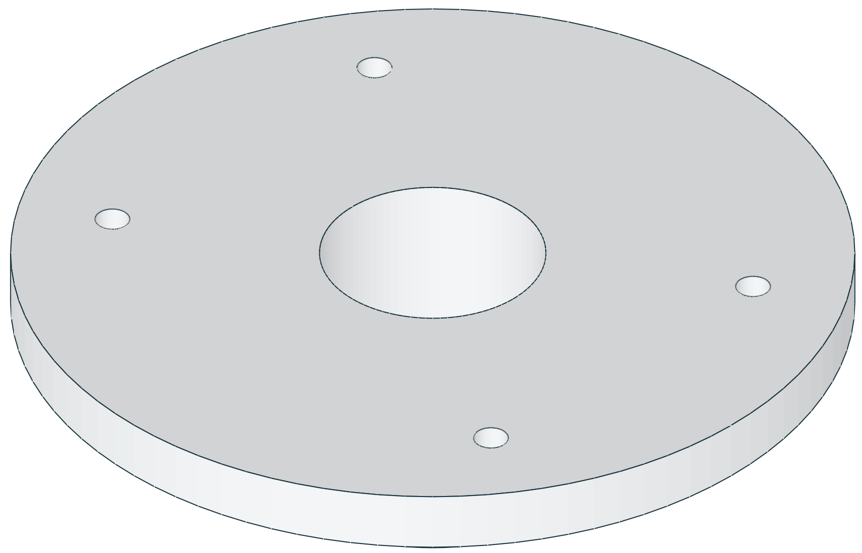
Bearing-supported yaw turntable for straight servo arm

turntable.stl | PRINT QUANTITY 1 | PETG

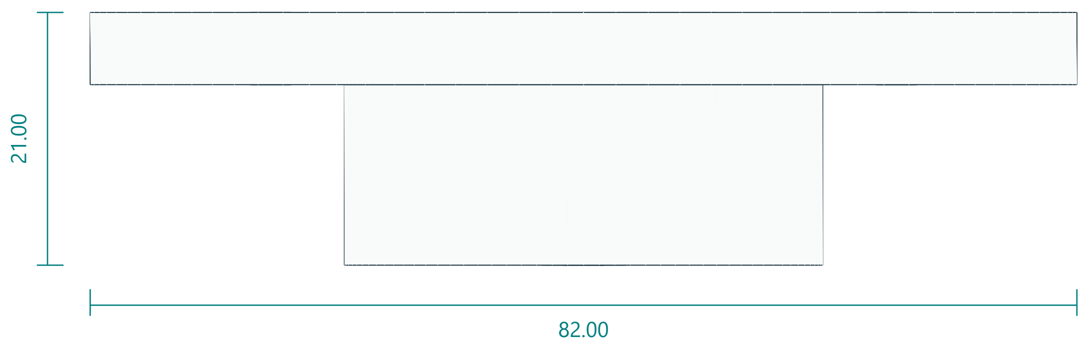
TOP / XY



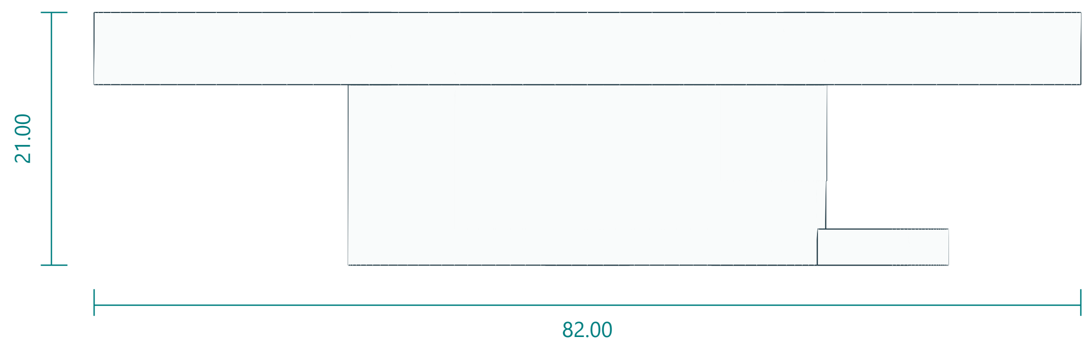
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

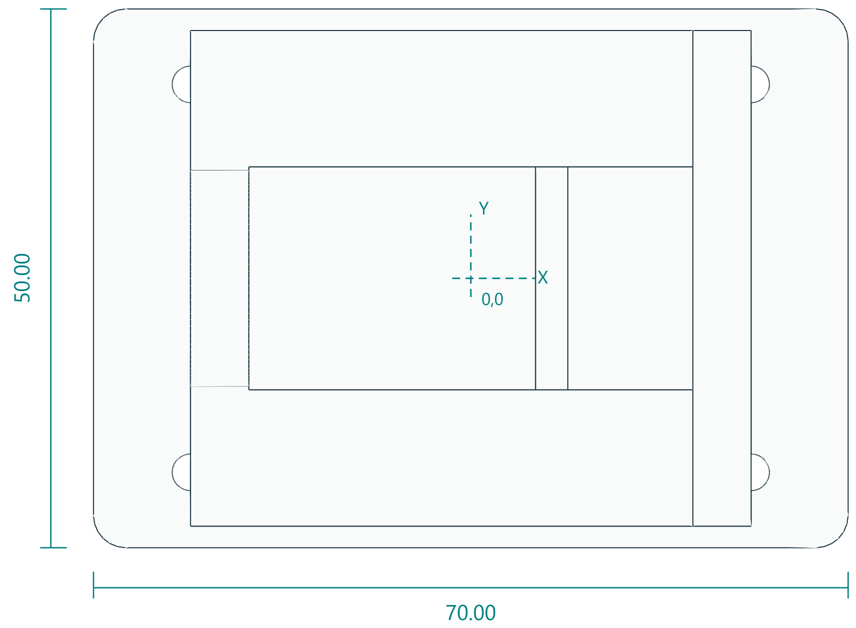


Interfaces: 82 diameter; neck39.75 x14.2; two adjustable radial horn slots; tower52 x36.
Fasteners / retention: 2 horn-arm bolts; 4 M3x16 tower.
Print orientation: neck down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Straight-horn center clearance7 with radial3.4 slots12-18 and20-25 in3mm web. Tower4x3.4 at(+/-26,+/-18). Top central tool aperture22.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

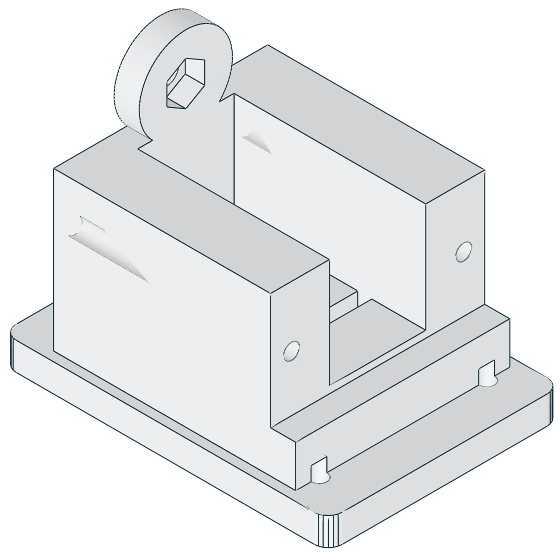
DS3218MG shoulder body tower with idler support

shoulder_tower.stl | PRINT QUANTITY 1 | PETG

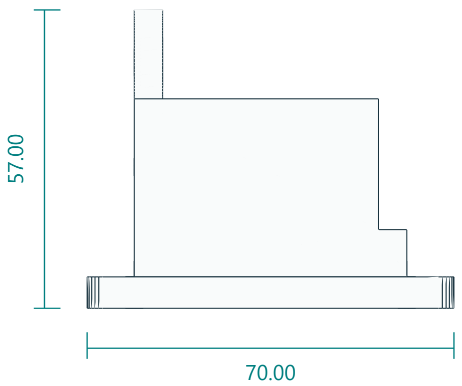
TOP / XY



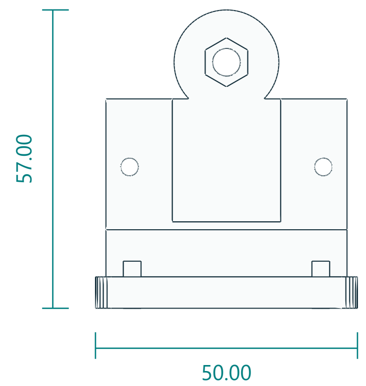
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

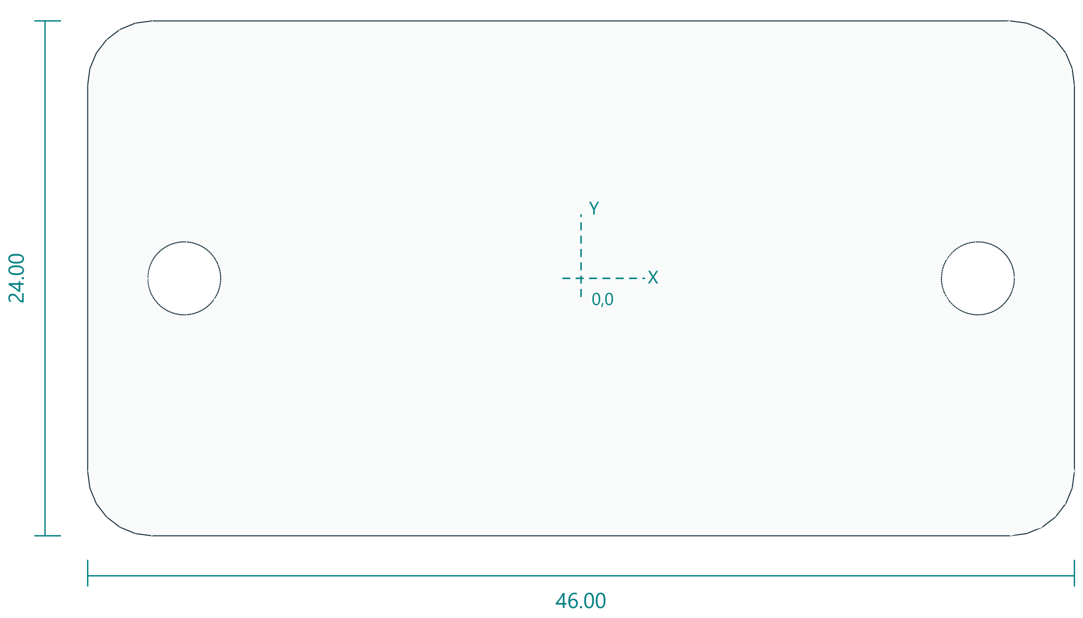


Interfaces: 70 x50; pitch axis47 above foot; body pocket41.2 x20.7; M5 idler nut trap.
Fasteners / retention: 4 M3x16; 2 M3x50 clamp; 1 M5 idler bolt and 625 bearing.
Print orientation: foot down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Foot4x3.4 at(+/-26,+/-18). Clamp holes alongX atY+/-18.5,Z29. Pitch axis47 abovefoot. Left idler M5 clearance and captive nut in20mm boss.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

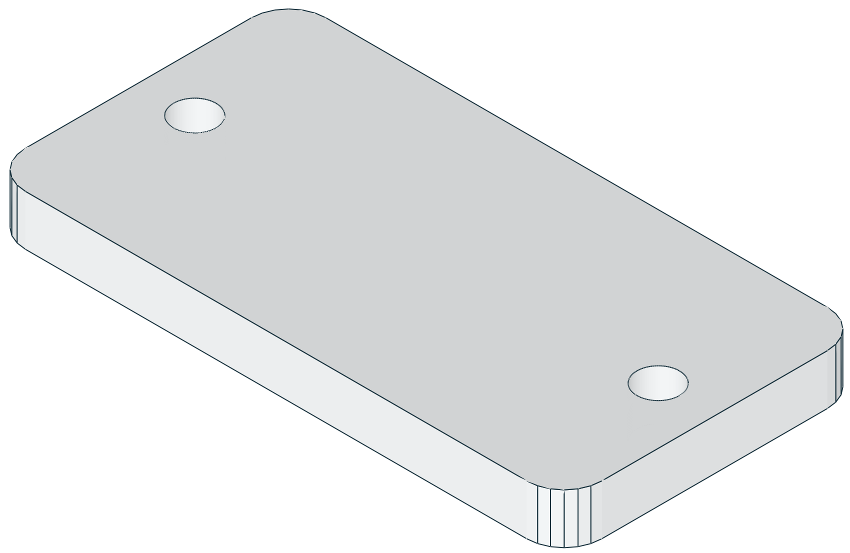
Shoulder clamp keeper

shoulder_cap.stl | PRINT QUANTITY 1 | PETG

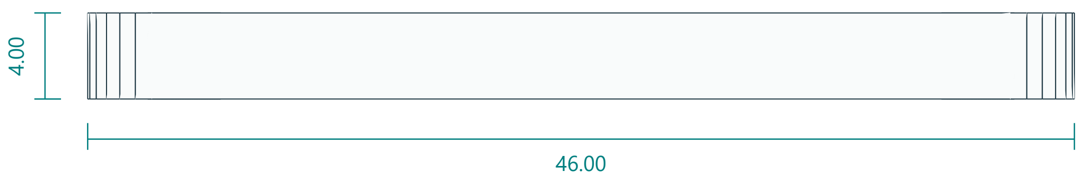
TOP / XY



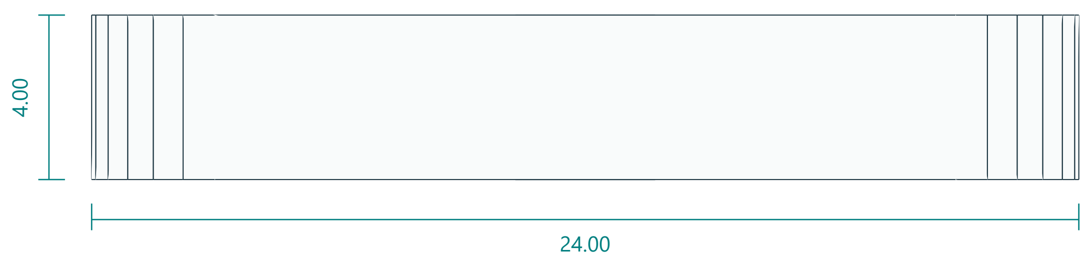
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 46 x28 x4; clamp pitch37.
Fasteners / retention: 2 M3x50.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,0). Printedface46x28.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

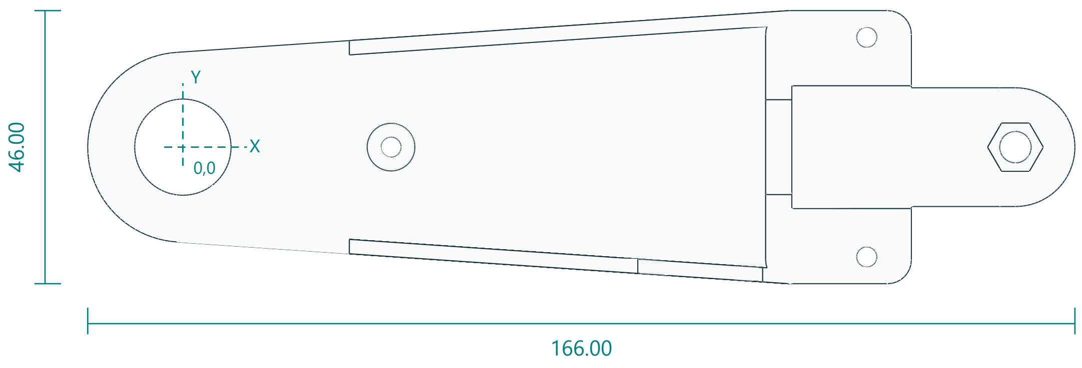
P08

REV C | 07 SEP 2026 | A3

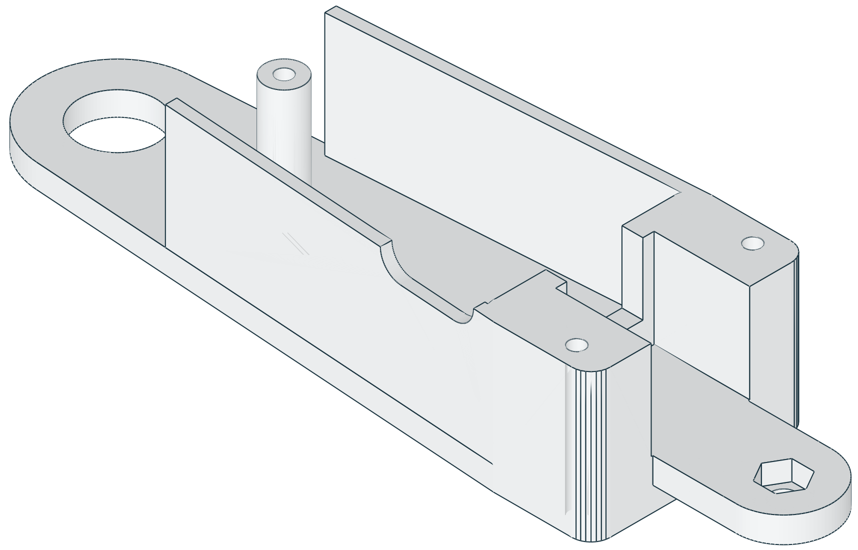
Upper arm enclosed link, idler half

upper_arm_left.stl | PRINT QUANTITY 1 | PETG

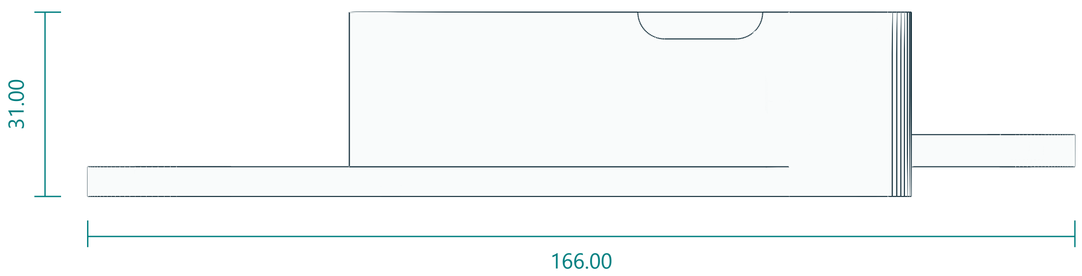
TOP / XY



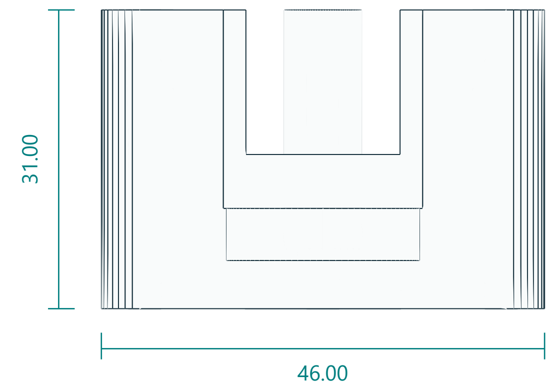
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 140 pivot pitch; 62 assembled width; 5 plate; 2.4 walls26 deep; 625 bearing pocket; distal M5 idler nut trap.
Fasteners / retention: 1 M5 idler bolt; 3 M3x70 through bolts shared with right half.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal 625 pocket16.2. Through bolt3.4 at(35,0); block bolts at(115,+/-18.5). DS3218 pocket41.2x20.7; walls fromX28; distal M5 nut trap atX140; cable port21x9 centredX87.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

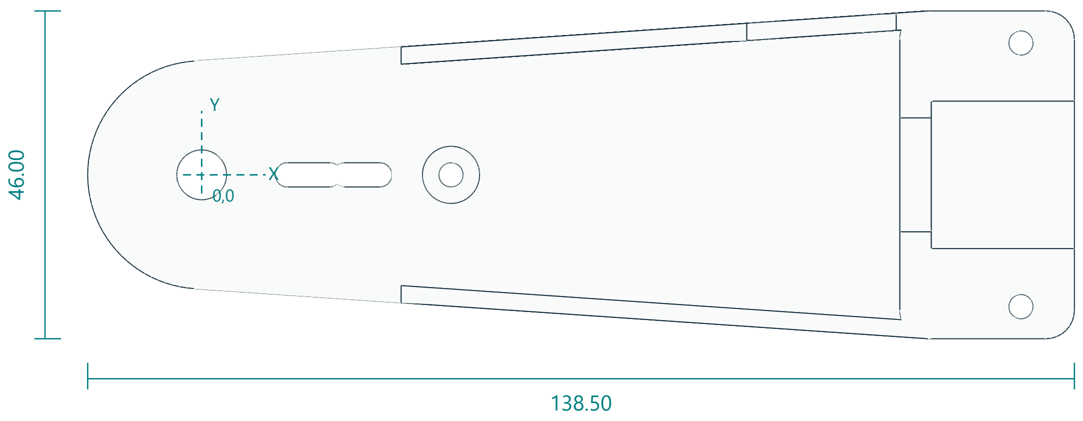
P09

REV C | 07 SEP 2026 | A3

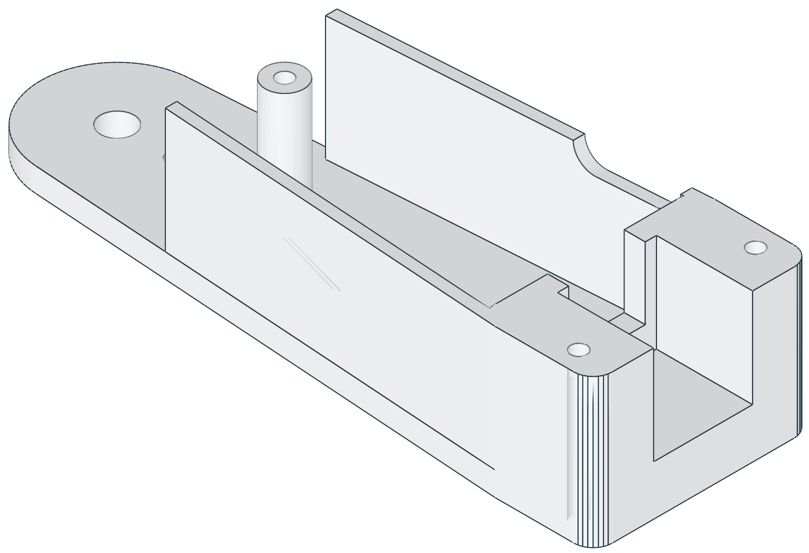
Upper arm enclosed link, driven half

upper_arm_right.stl | PRINT QUANTITY 1 | PETG

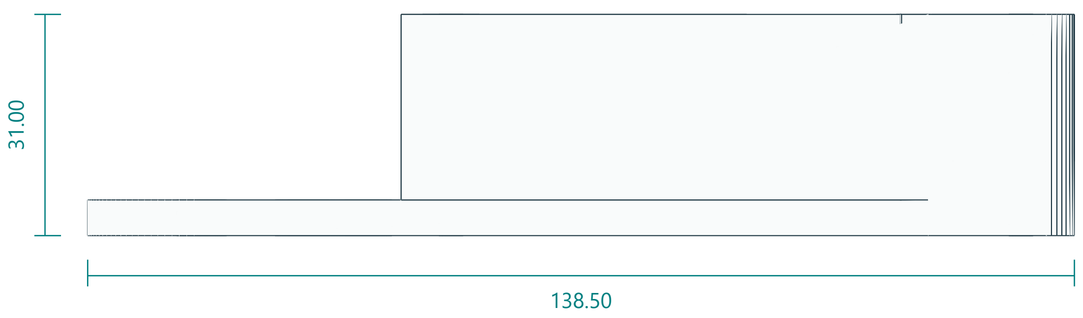
TOP / XY



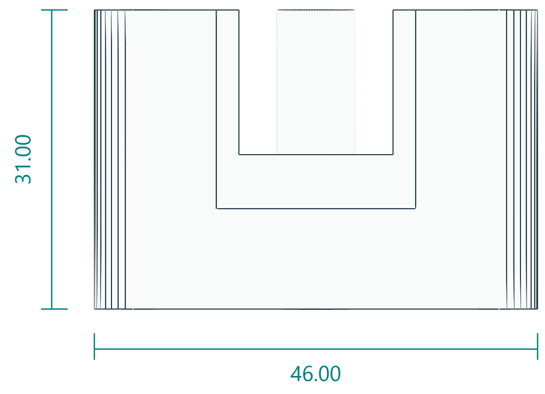
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 140 pivot pitch; 62 assembled width; two proximal horn-arm slots; cable port on inner elbow wall.
Fasteners / retention: 2 horn-arm bolts; shared M3x70.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Driven side has center7 and radial3.4 slots12-18 and20-25. Same clamp-bolt pattern as left; cable port remains on the inner elbow wall after assembly.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

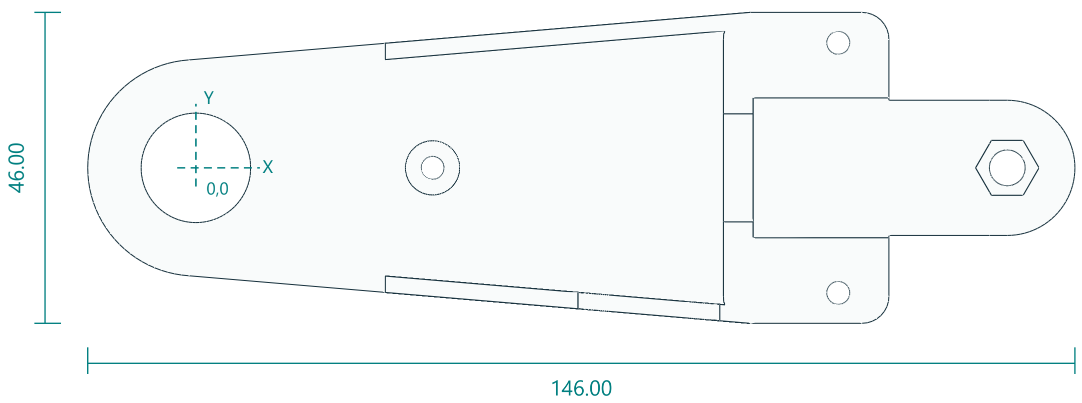
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

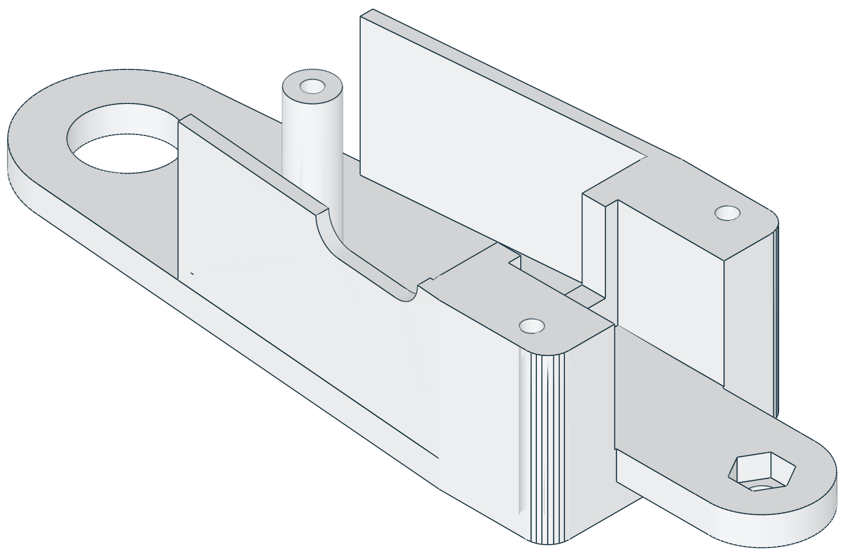
Forearm enclosed link, idler half

forearm_left.stl | PRINT QUANTITY 1 | PETG

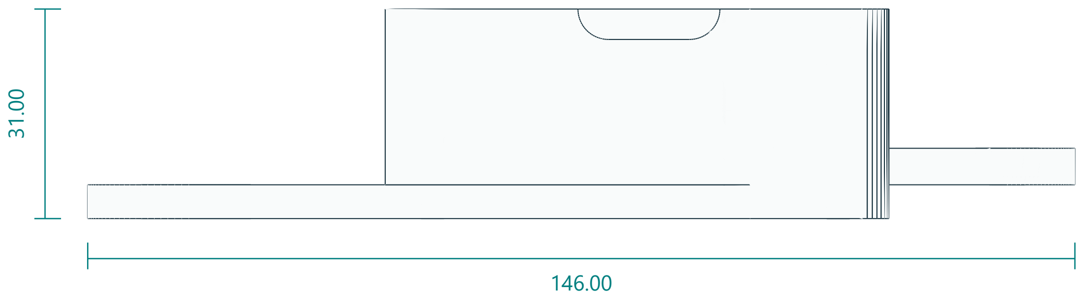
TOP / XY



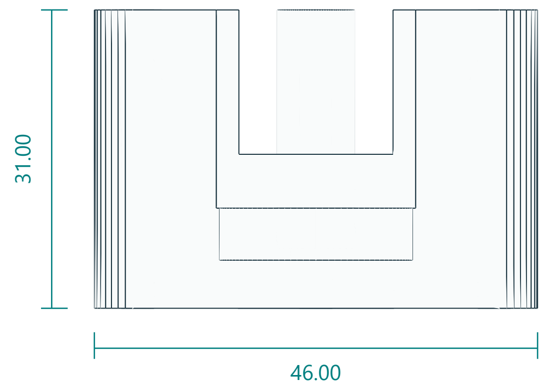
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

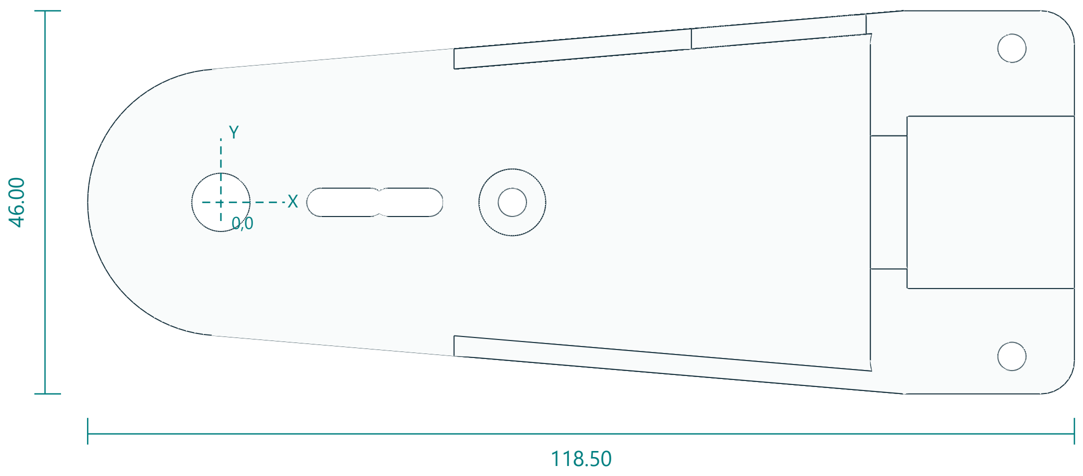


Interfaces: 120 pivot pitch; 62 assembled width; 625 bearing pocket; distal M5 idler nut trap.
Fasteners / retention: 1 M5 idler bolt; 3 M3x70 through bolts shared with right half.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal 625 pocket16.2. Through bolt3.4 at(35,0); block bolts at(95,+/-18.5). DS3218 pocket41.2x20.7; distal M5 nut trap atX120; cable port21x9 centredX67.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

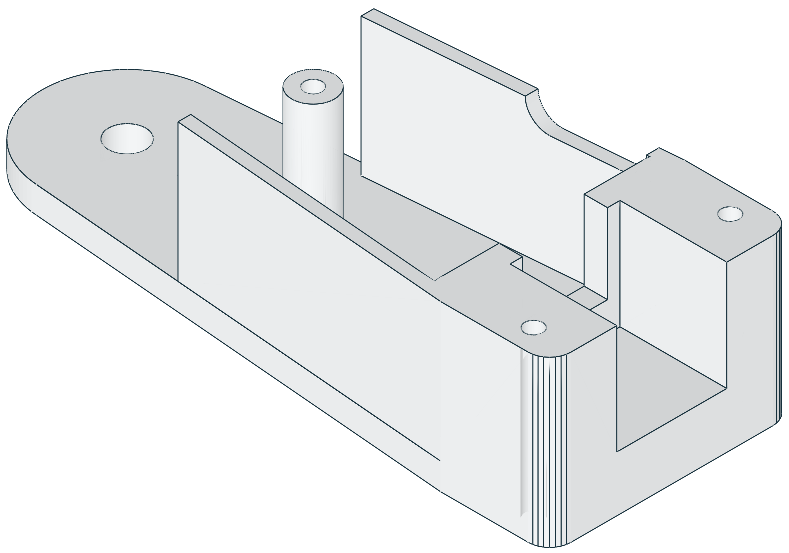
Forearm enclosed link, driven half

forearm_right.stl | PRINT QUANTITY 1 | PETG

TOP / XY

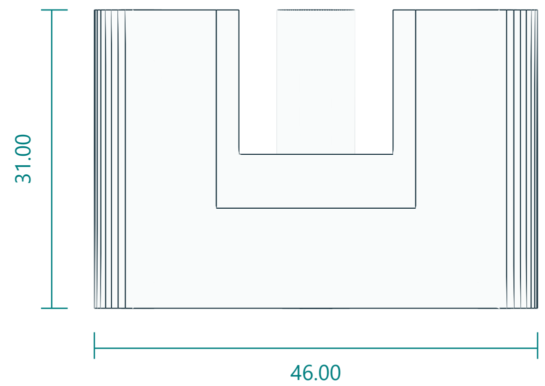


ISOMETRIC / REFERENCE



FRONT / XZ

RIGHT / YZ



Interfaces: 120 pivot pitch; 62 assembled width; two proximal horn-arm slots; cable port on inner wrist wall.
Fasteners / retention: 2 horn-arm bolts; shared M3x70.
Print orientation: plate down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Driven side has center7 and radial3.4 slots12-18 and20-25. Same clamp-bolt pattern as left; cable port remains on the inner wrist wall after assembly.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

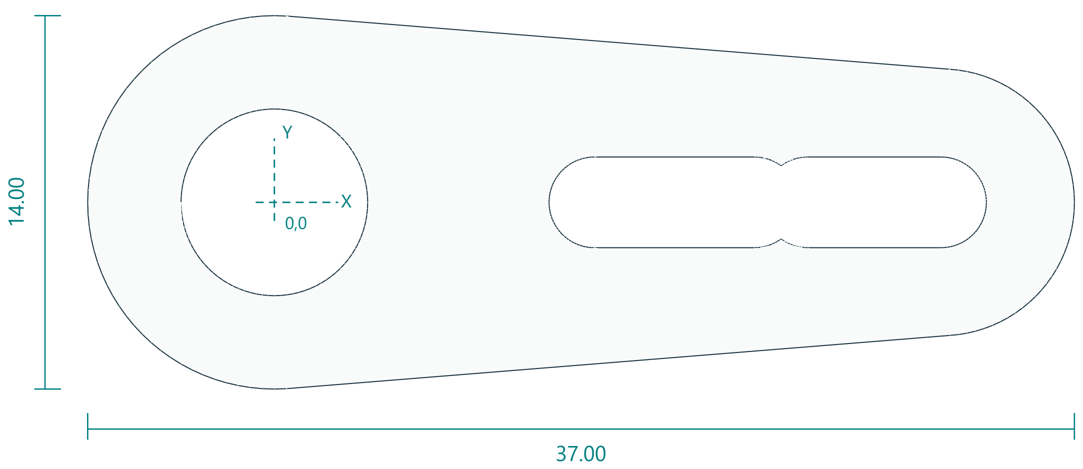
P12

REV C | 07 SEP 2026 | A3

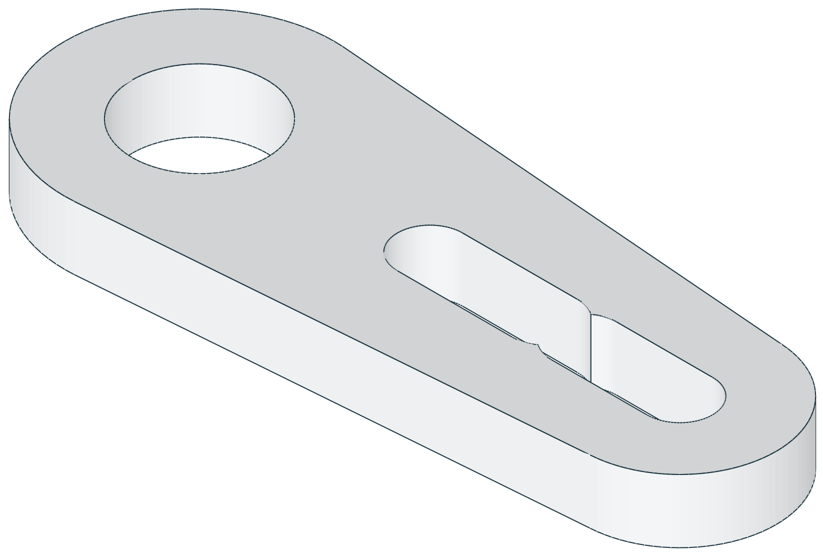
DS3218 straight-horn to driven-link spacer

horn_spacer.stl | PRINT QUANTITY 3 | PETG

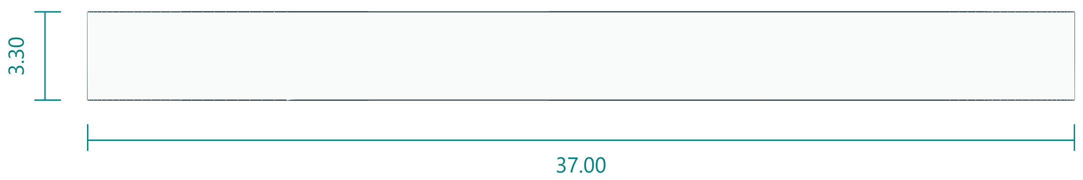
TOP / XY



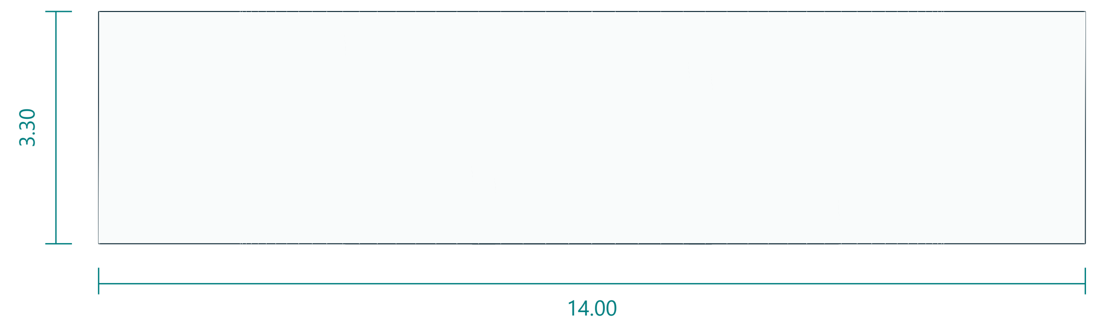
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 30 x14 x3.3; center7; adjustable radial slots12-18 and20-25.
Fasteners / retention: 2 horn-arm bolts each.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Center7; radial3.4 slots12-18 and20-25; thickness3.3; one driven-side spacer per pitch joint.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

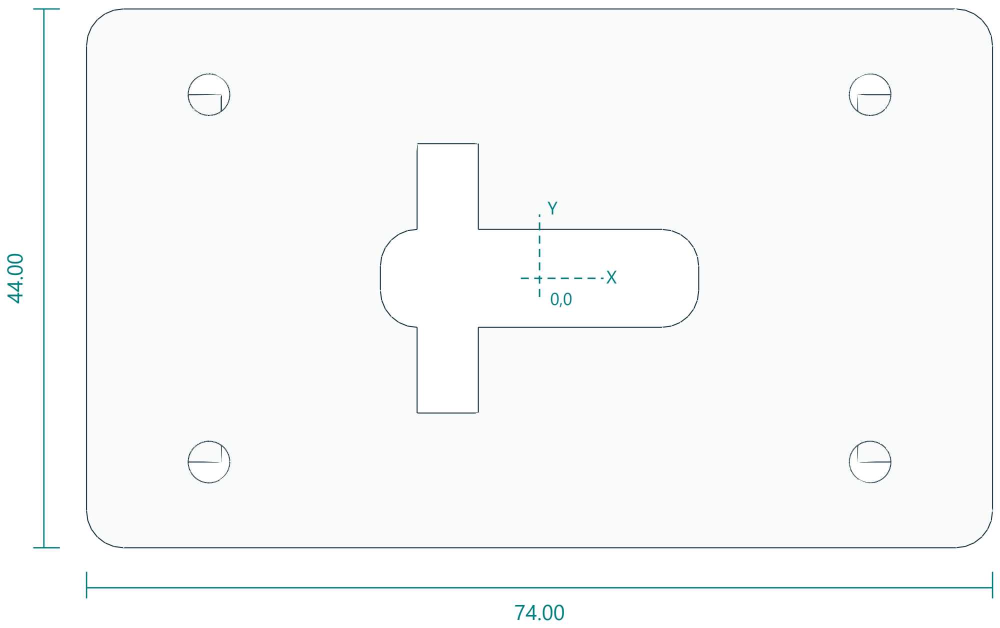
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

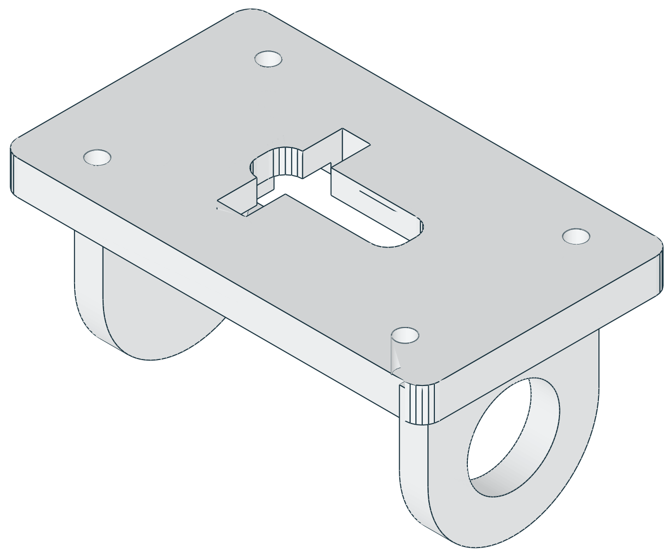
Head tilt fork with driven and 625-idler ears

head_yoke.stl | PRINT QUANTITY 1 | PETG

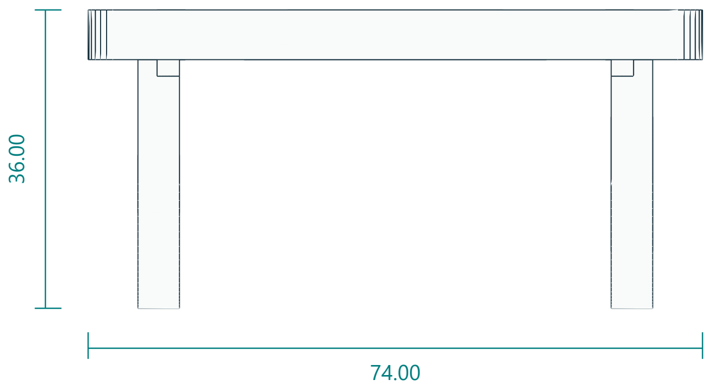
TOP / XY



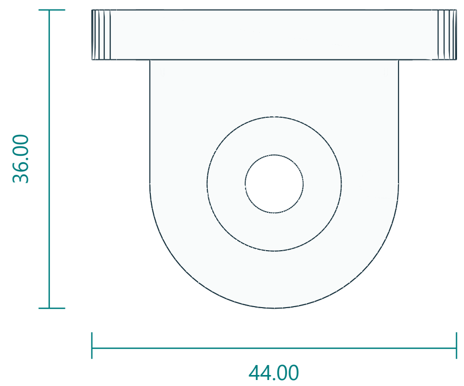
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 62 fork inside span; axis21 behind head rear; rear holes54 x30; 26 x8 centre cable slot.
Fasteners / retention: 2 horn-arm bolts; 1 M5 idler bolt; 1 625 bearing; 4 M3x12 rear screws.
Print orientation: rear mounting plate on bed; rotate180X. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Back bolts3.4 XY(+/-27,+/-15); attachmentplaneZ36. AxisX atY0,Z15; one625 pocket16.2 and one tangential straight-horn interface; fork inside width52. Centre cable slot26x8.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

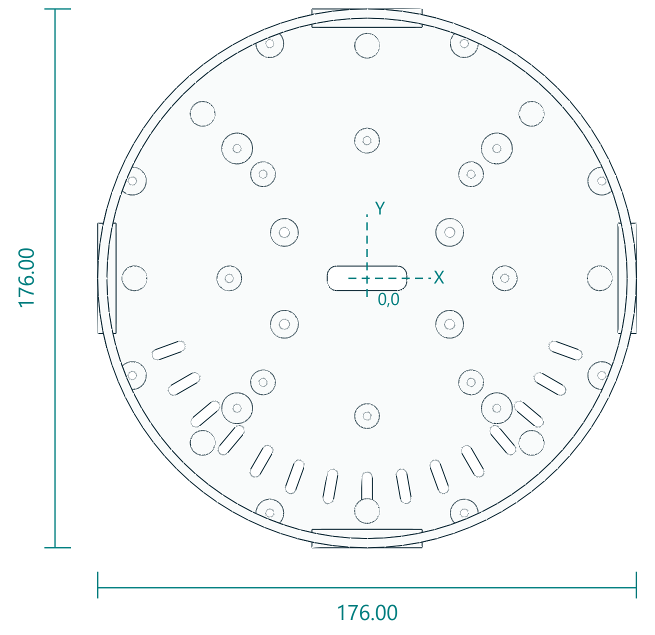
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

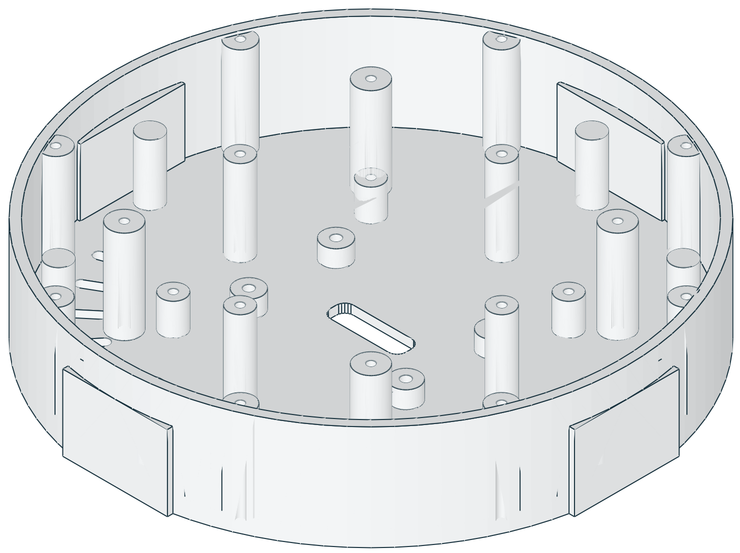
Optical head rear enclosure

head_shell.stl | PRINT QUANTITY 1 | PETG

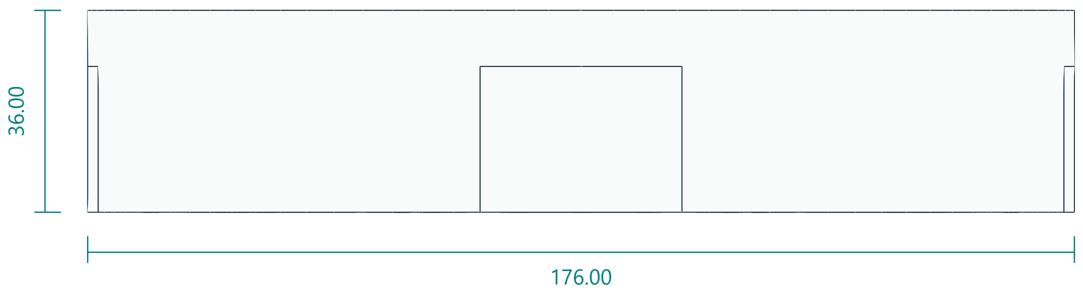
TOP / XY



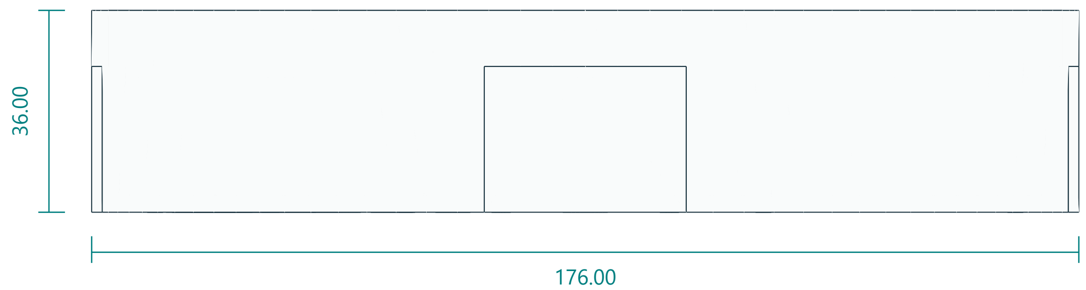
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 176 diameter x36; rear thickness3; white carrier bosses radius45 top32.5; 26 x8 centre cable slot.
Fasteners / retention: 8 M3x10 bezel; 4 M3x12 yoke.
Print orientation: rear disc down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Rear bolts(+/-27,+/-15); LCDpilots(+/-34,+/-34),top14. Whitecarrierpilotsradius45 cardinal,top32.5. Facebossradius60 angles45+90n,top35. Bezelradius83 angles22.5+45n,top35. Centre cable slot26x8. No sensor penetrations.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

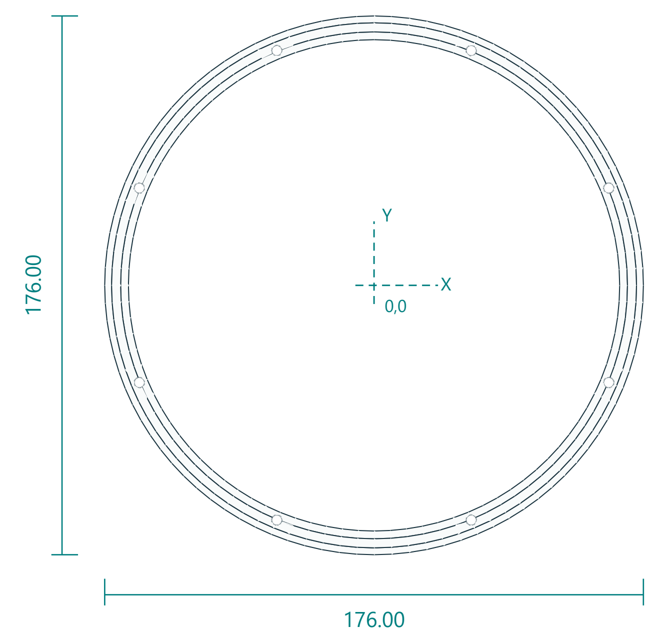
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

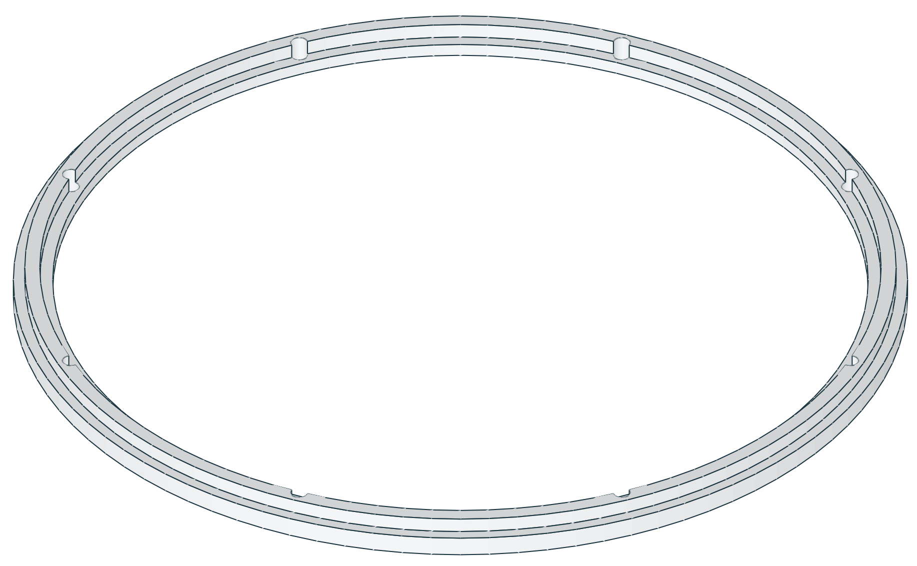
Outer RGB diffuser retaining bezel

outer_bezel.stl | PRINT QUANTITY 1 | PETG

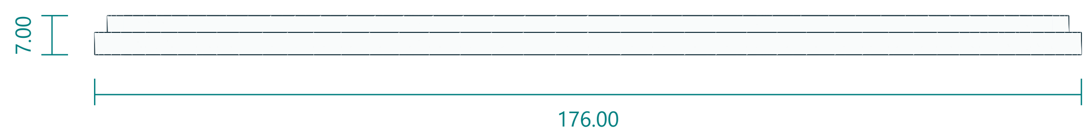
TOP / XY



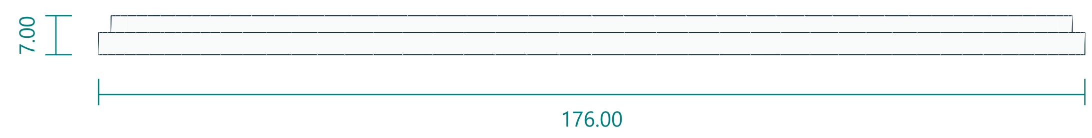
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 176 diameter; window160.5; 8 holes radius83.
Fasteners / retention: 8 M3x10.
Print orientation: front lip down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Eight3.4 holes radius83 angles22.5+45n. Rearflangerecessdiameter165.5 by1.4 deep; facewindow160.5.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

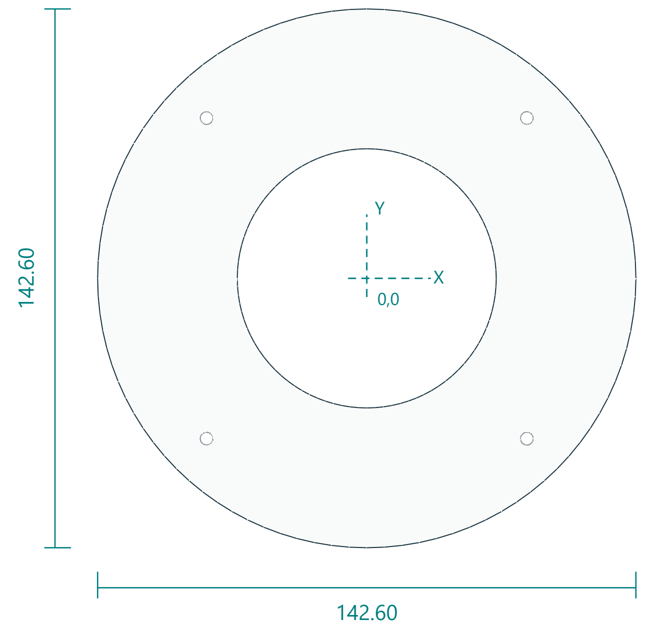
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

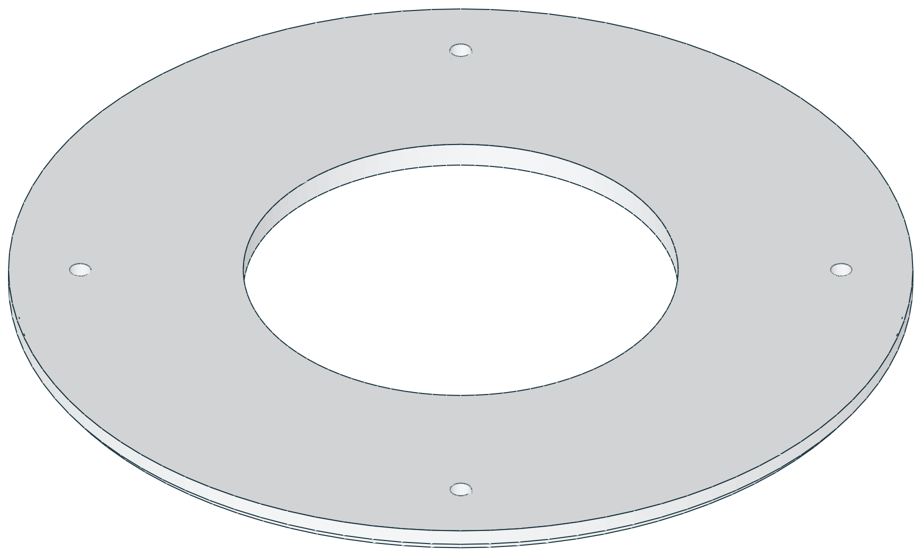
Central optical baffle plate

face_center.stl | PRINT QUANTITY 1 | black PETG

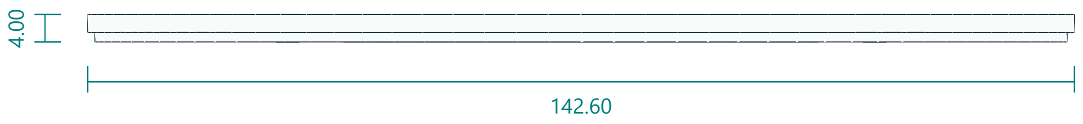
TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

Interfaces: 142.6 diameter x4; center68.6; screws radius60.
Fasteners / retention: 4 M3x12 into shell bosses.
Print orientation: flat front down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holesradius60 angles45+90n. No sensor penetrations.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

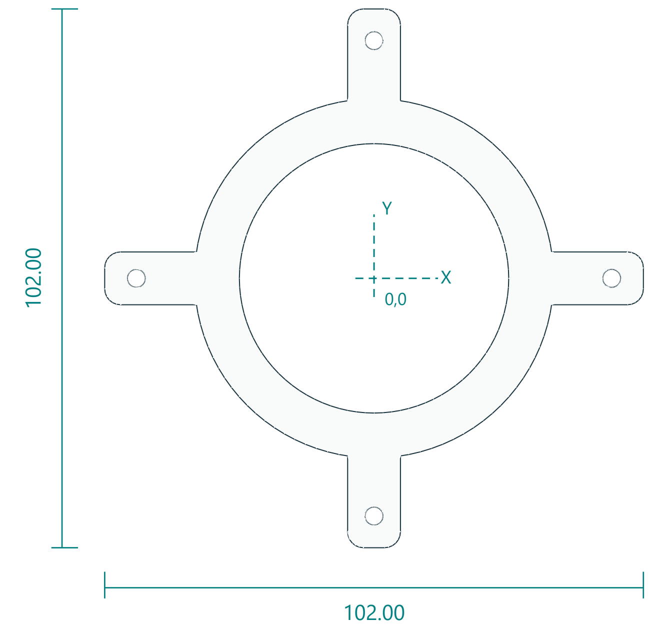
UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

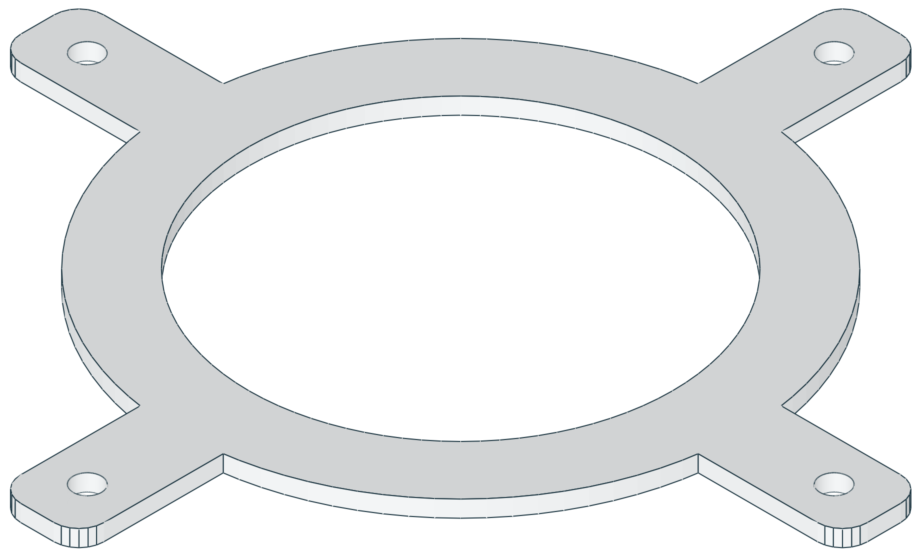
White ring carrier and mounting arms

white_carrier.stl | PRINT QUANTITY 1 | PETG

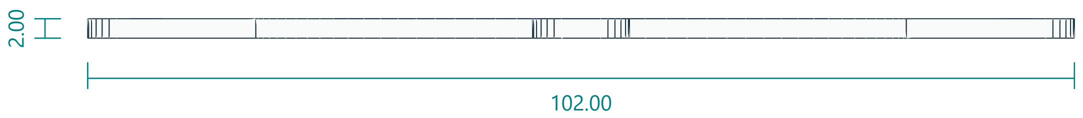
TOP / XY



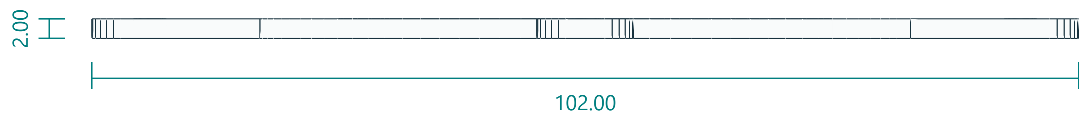
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

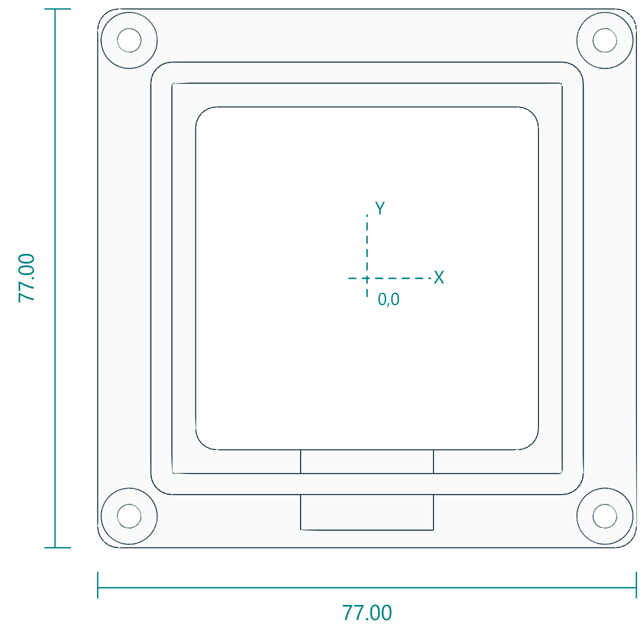


Interfaces: 68 OD51 ID; 4 cardinal mounting holes radius45; 2 thick.
Fasteners / retention: 4 M3x10.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes at(+/-45,0),(0,+/-45). Allarmsandring2mmthick.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

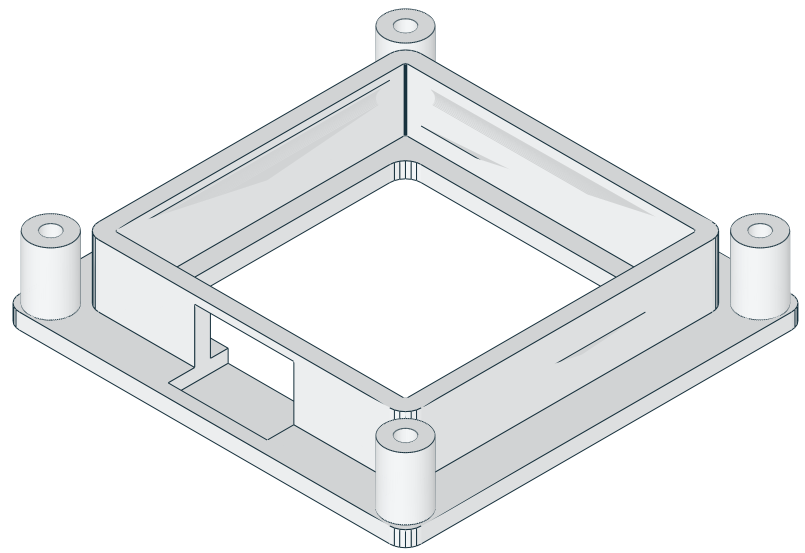
55 mm LCD edge cradle

lcd_cradle.stl | PRINT QUANTITY 1 | PETG

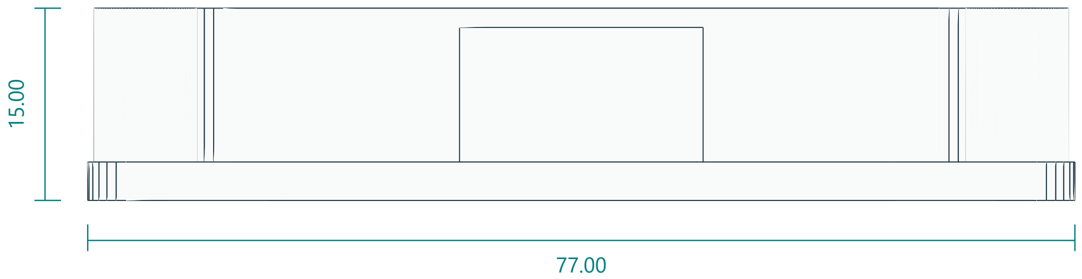
TOP / XY



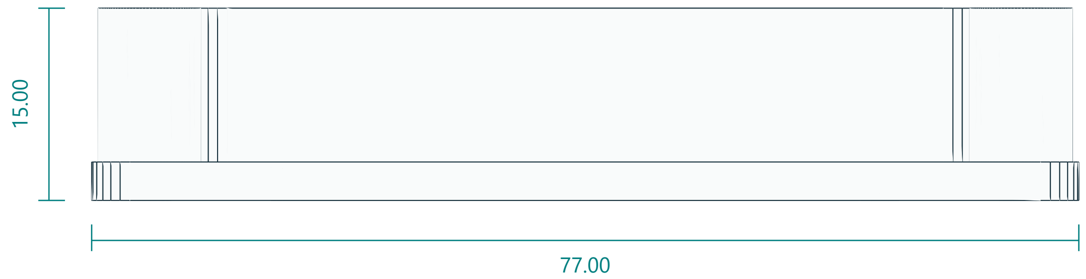
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

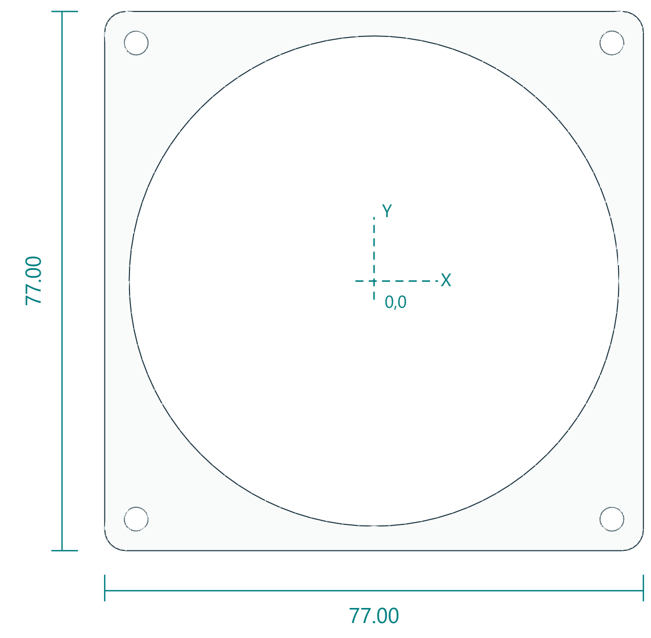


Interfaces: 55.8 square pocket; 15 depth; USB19 x13 relief.
Fasteners / retention: 4 M3x25; thin closed cell foam shims.
Print orientation: flat base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Inner55.8square cavity. Rear49squareaccess. USBopening19wide at-Y.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

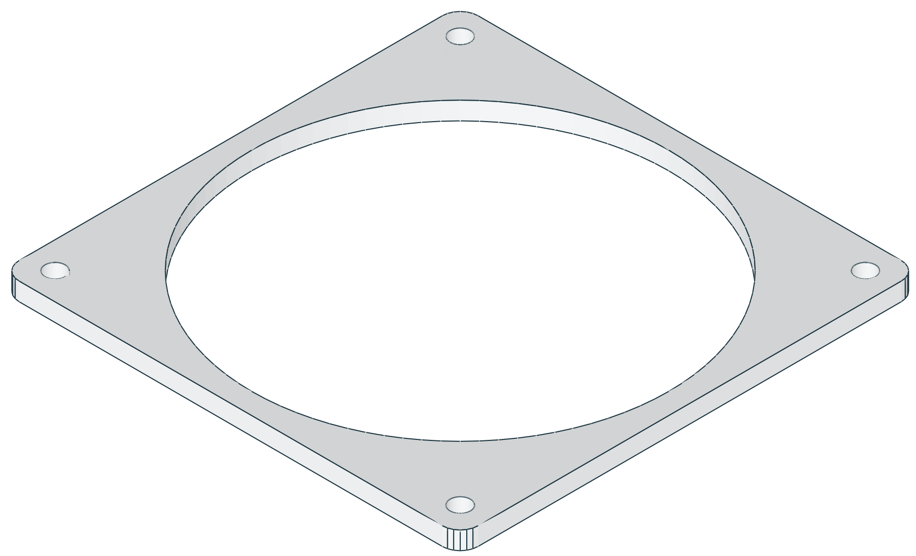
LCD PCB corner retaining plate

lcd_retainer.stl | PRINT QUANTITY 1 | black PETG

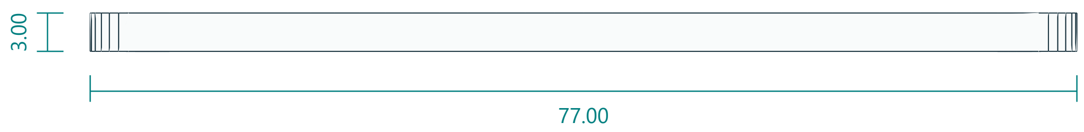
TOP / XY



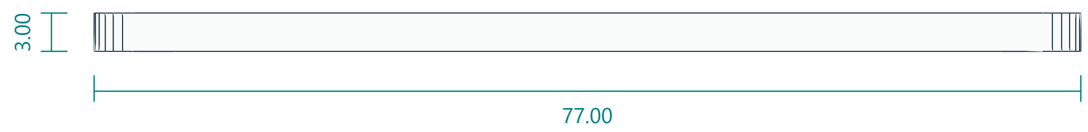
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

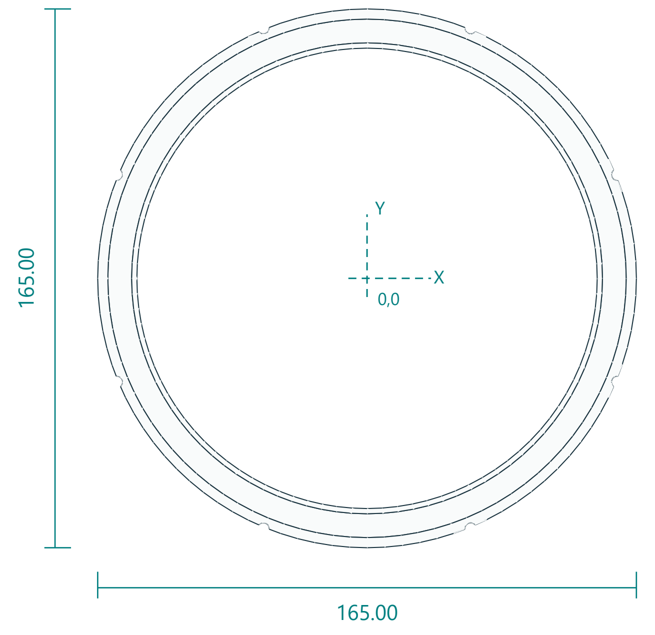


Interfaces: 77 square x3; opening70; mounting pitch68.
Fasteners / retention: 4 M3x25 shared.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Diameter70centeropening retainsPCBcorners, not roundglass.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

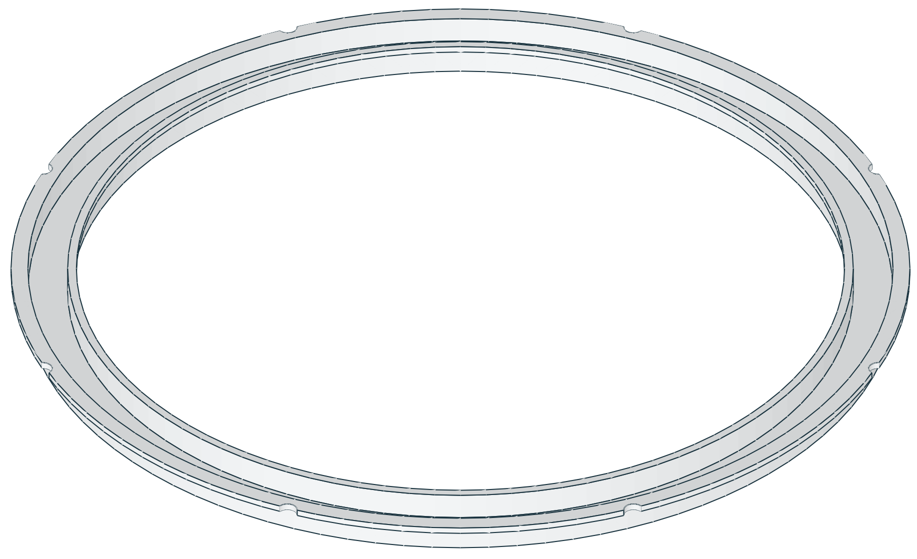
Frosted outer RGB annular diffuser

outer_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

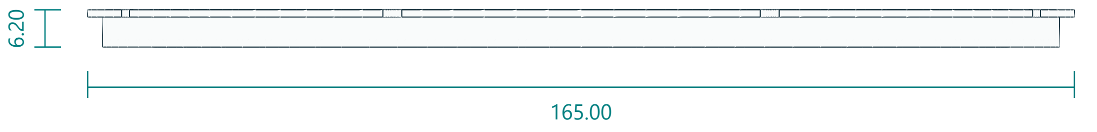
TOP / XY



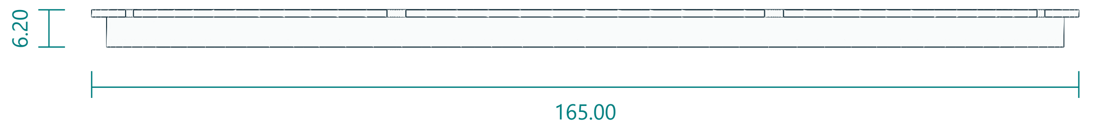
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 160 face OD143 ID; 165 flange; 1.2 face; 6.2 depth.
Fasteners / retention: captured by outer bezel/center plate.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: FaceOD160 ID143. RearflangesOD165/ID158.8 andOD144.2/ID141. M3edgeclearancenotchesradius83.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

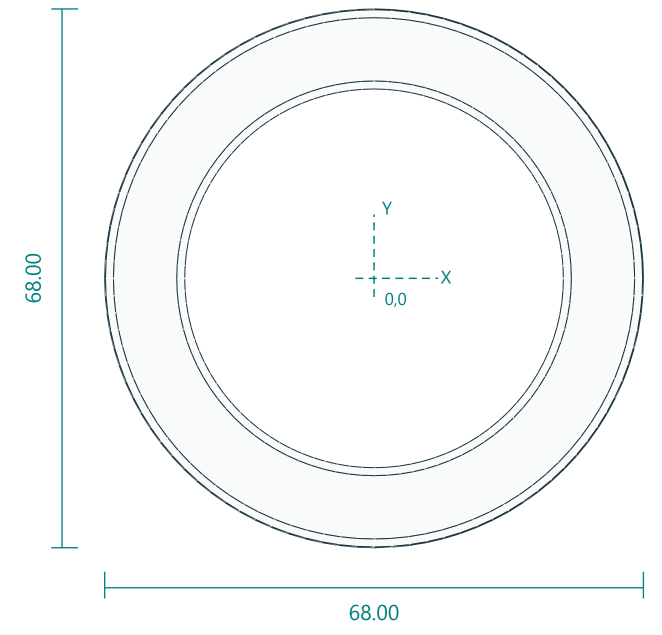
P21

REV C | 07 SEP 2026 | A3

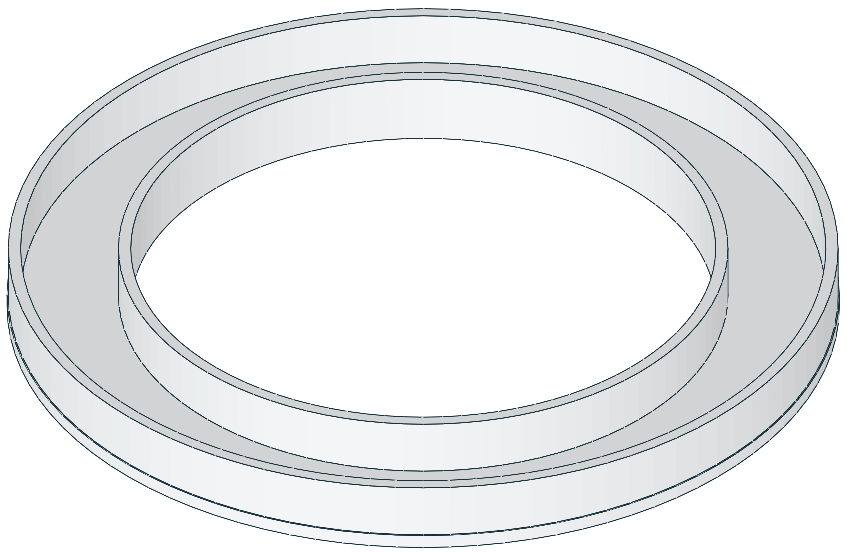
Frosted inner RGBW light diffuser

inner_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

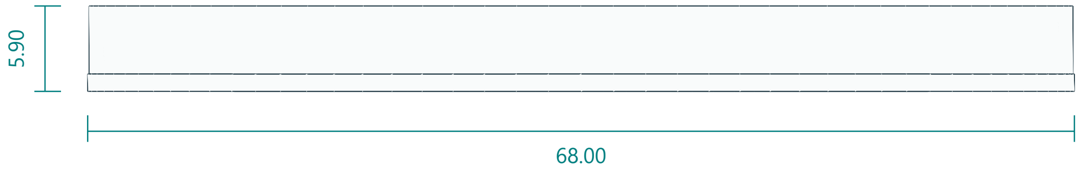
TOP / XY



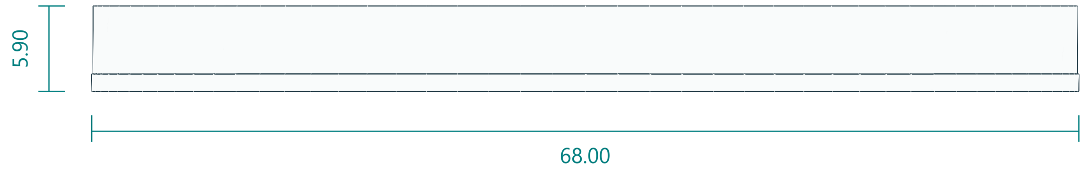
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 68 OD47.8 ID; 1.2 face; 5.9 depth.
Fasteners / retention: 3 tiny neutral cure silicone retention dots.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: OD68 ID47.8; skin1.2; depth5.9. No sensorwindowmaterial.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

Views fitted to sheet. Do not measure this PDF. Digital prototype; check physical fit before batch printing.

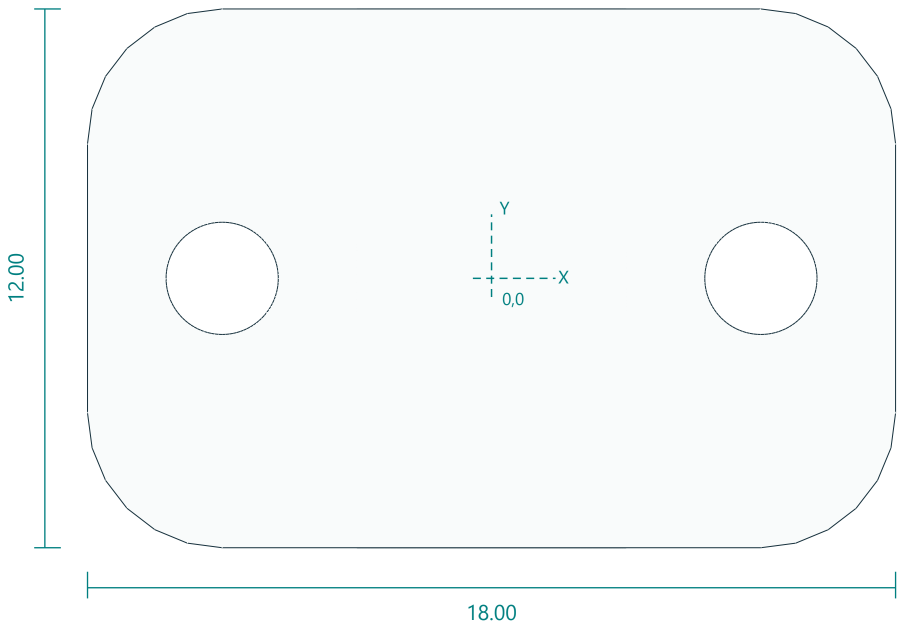
P22

REV C | 07 SEP 2026 | A3

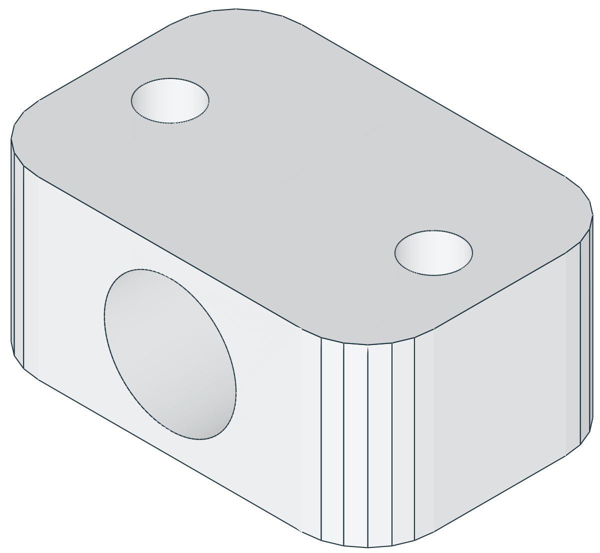
Harness saddle clip

cable_clip.stl | PRINT QUANTITY 6 | PETG

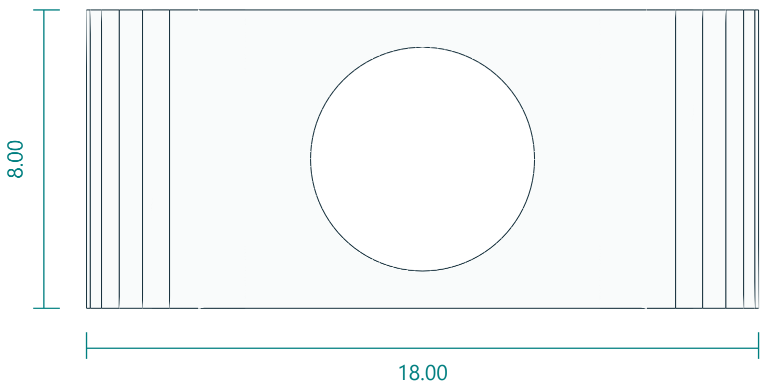
TOP / XY



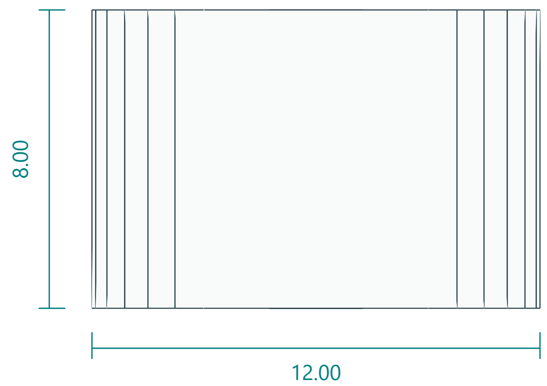
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ



Interfaces: 18 x12 x8; cable6; screws12 pitch.
Fasteners / retention: 2 M2.5x12 each optional.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two2.5holes(+/-6,0); cablebore6 alongY centeredZ4.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

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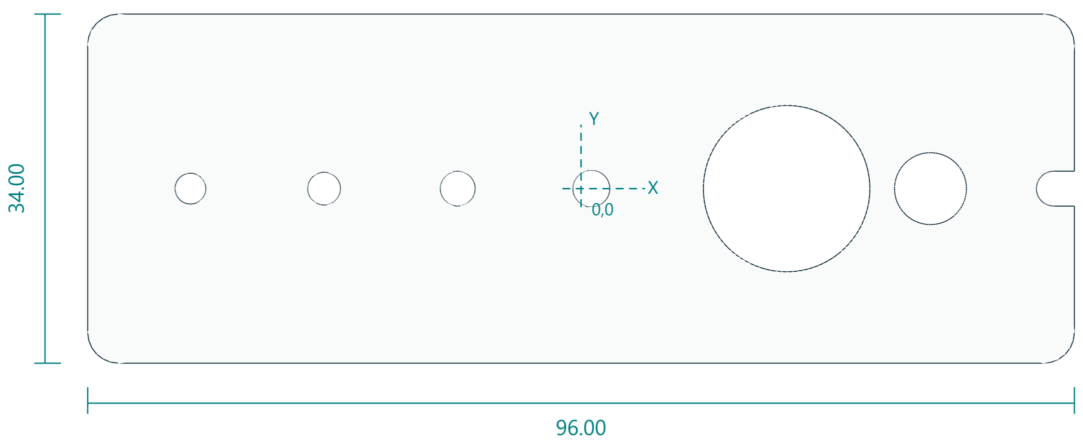
P23

REV C | 07 SEP 2026 | A3

Fastener, 625 bearing and horn-slot clearance coupon

fit_coupon.stl | PRINT QUANTITY 1 | PETG

TOP / XY



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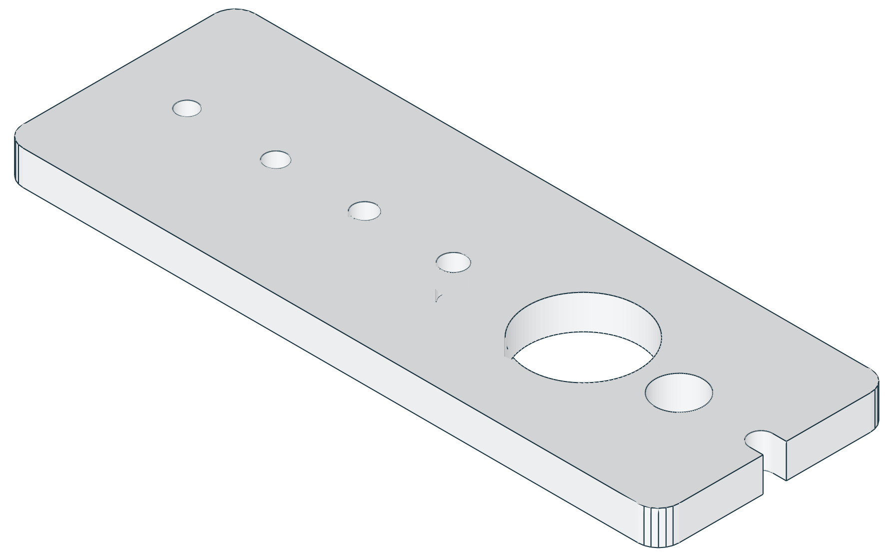
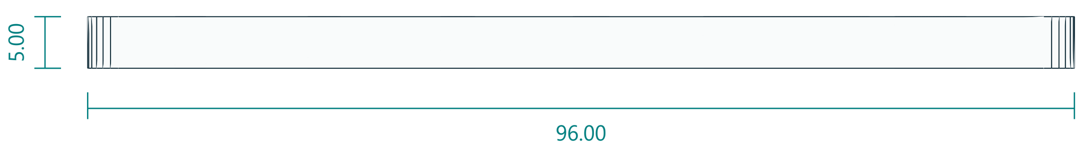
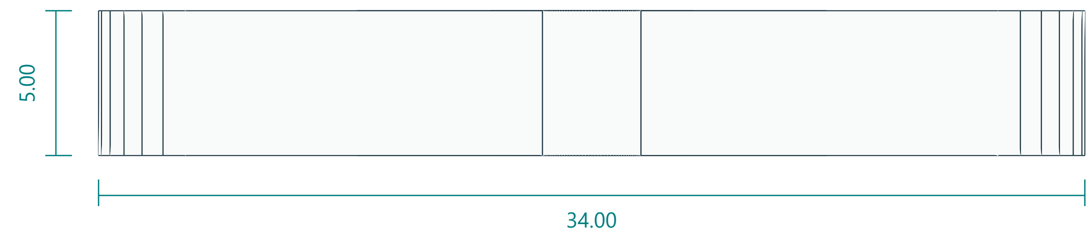


FRONT / XZ



RIGHT / YZ



Interfaces: 3.0 /3.2 /3.4 /3.6 bores; 16.2 bearing pocket; DS3218 horn slots.
Fasteners / retention: none.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolt bores3.0,3.2,3.4,3.6; 625 bearing pocket16.2; straight-horn center7 and radial slots12-18 and20-25.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

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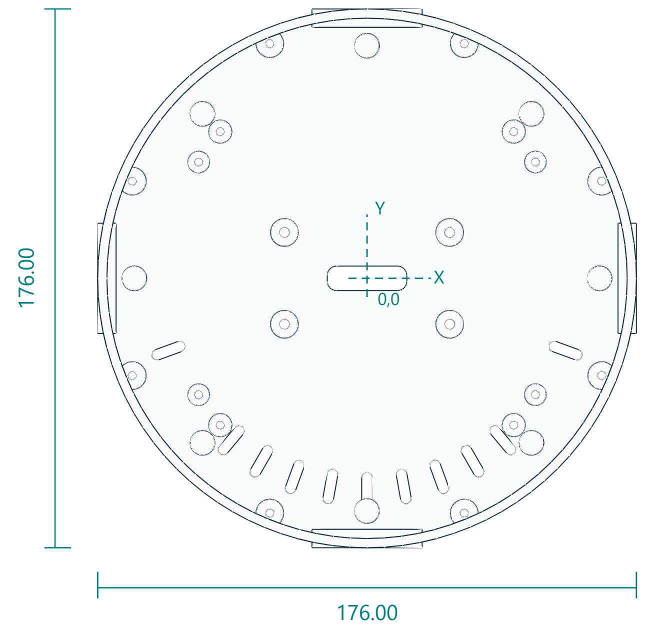
P24

REV C | 07 SEP 2026 | A3

Optional DSI head rear enclosure

head_shell_dsi.stl | PRINT QUANTITY 1 | PETG | OPTIONAL 4-INCH DSI HEAD ONLY

TOP / XY



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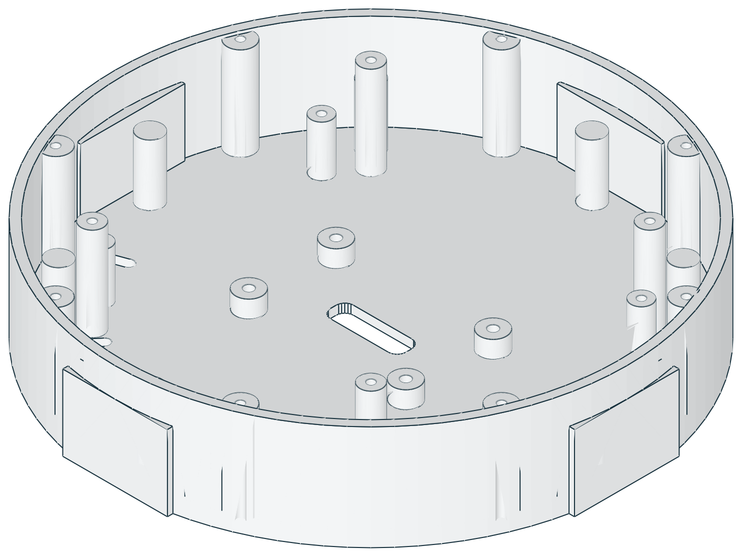
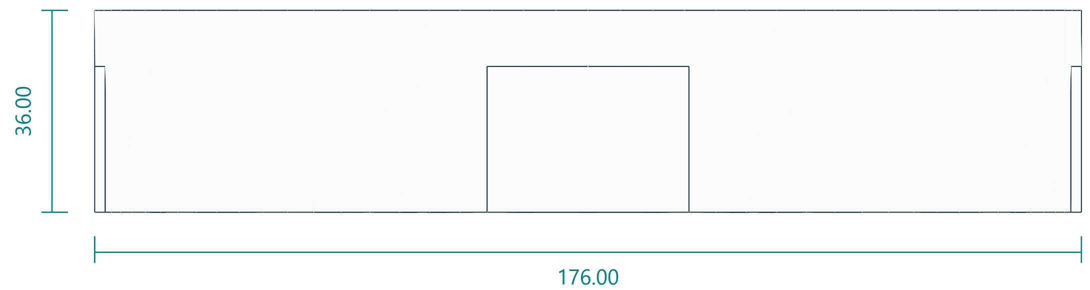
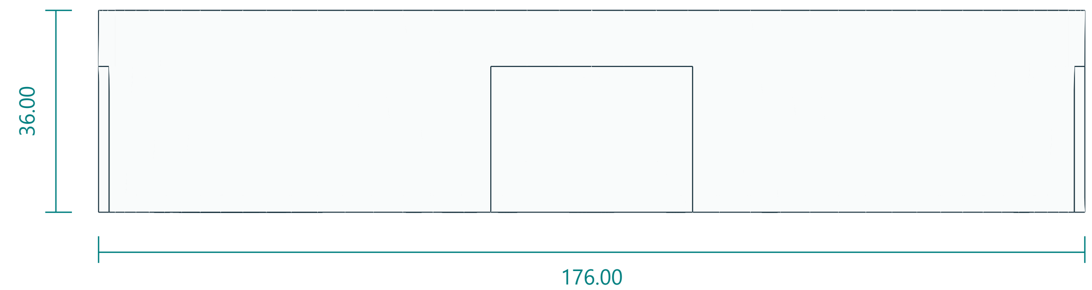


FRONT / XZ



RIGHT / YZ

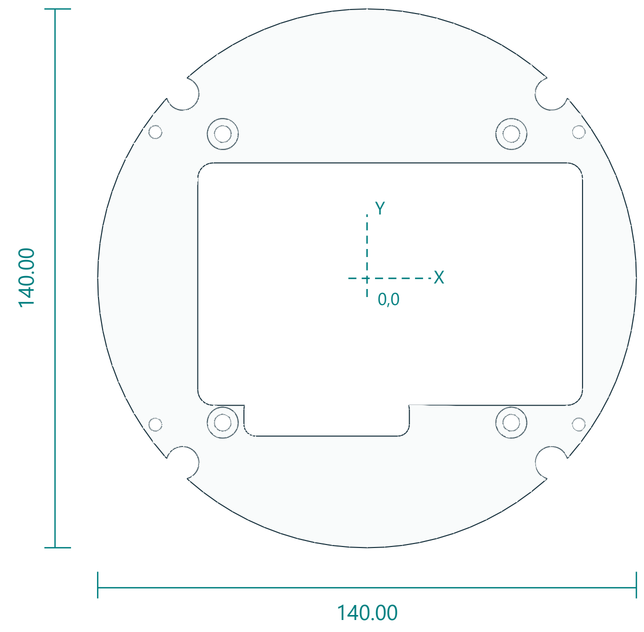


Interfaces: 176 diameter x36; carrier bosses (+/-55,+/-38) top20.5; face ring bosses radius67.8 top35.
Fasteners / retention: 8 M3x10 bezel; 4 M3x12 yoke; 4 M3x8 carrier; 4 M3x12 face ring.
Print orientation: rear disc down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Rear bolts(+/-27,+/-15). Carrier pilots2.8 at(+/-55,+/-38) top20.5. Face ring pilots2.8 radius67.8 angles45+90n top35. Bezel radius83 angles22.5+45n top35. Centre cable slot26x8. No sensor penetrations.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

Optional 4 inch LCD carrier plate

lcd4_carrier.stl | PRINT QUANTITY 1 | PETG | OPTIONAL 4-INCH DSI HEAD ONLY

TOP / XY



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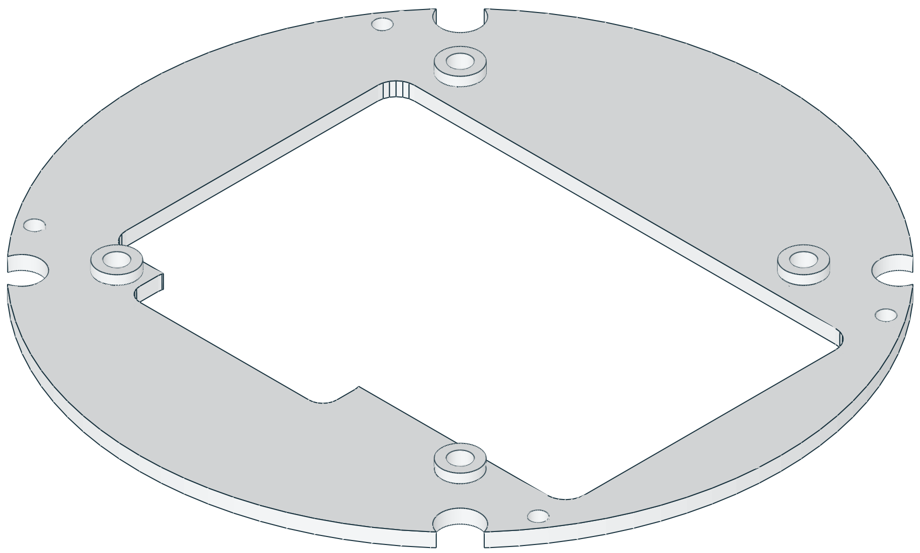
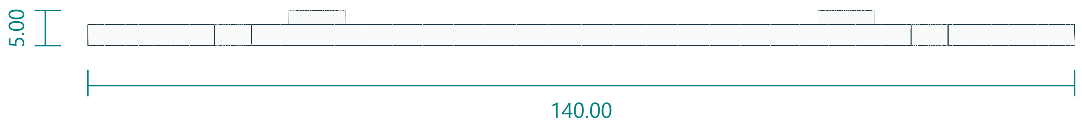
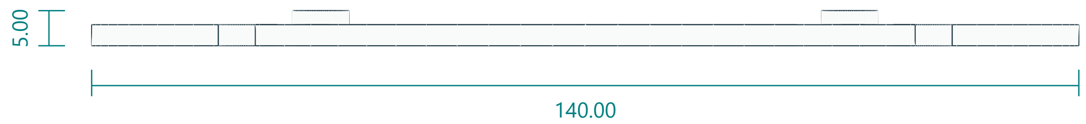


FRONT / XZ



RIGHT / YZ



Interfaces: 140 diameter x3; boss pads to5; LCD M4 pattern75 x75; PCB window100 x63 with connector notch.
Fasteners / retention: 4 M4x8 into LCD case bosses; 4 M3x8 into shell bosses.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 clearance4.4 at(+/-37.5,+/-37.5) on5mm pads. Shell screws3.4 at(+/-55,+/-38). Face ring boss clearance8.5 radius67.8. PCB window X-44..56 Y-33..30 plus connector notch X-32..11 Y-41..-31.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

UNITS mm · THIRD-ANGLE PART VIEWS · NUMERICAL DIMENSIONS CONTROL

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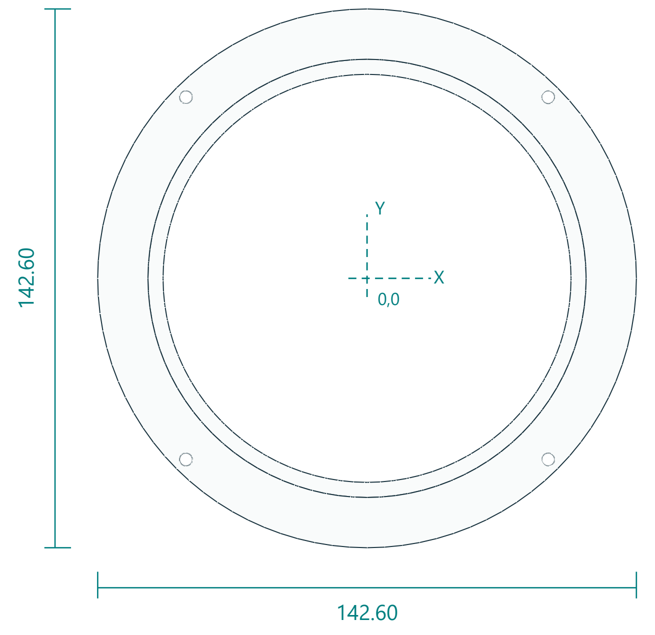
Optional DSI face ring

face_ring_dsi.stl | PRINT QUANTITY 1 | black PETG | OPTIONAL 4-INCH DSI HEAD ONLY

TOP / XY



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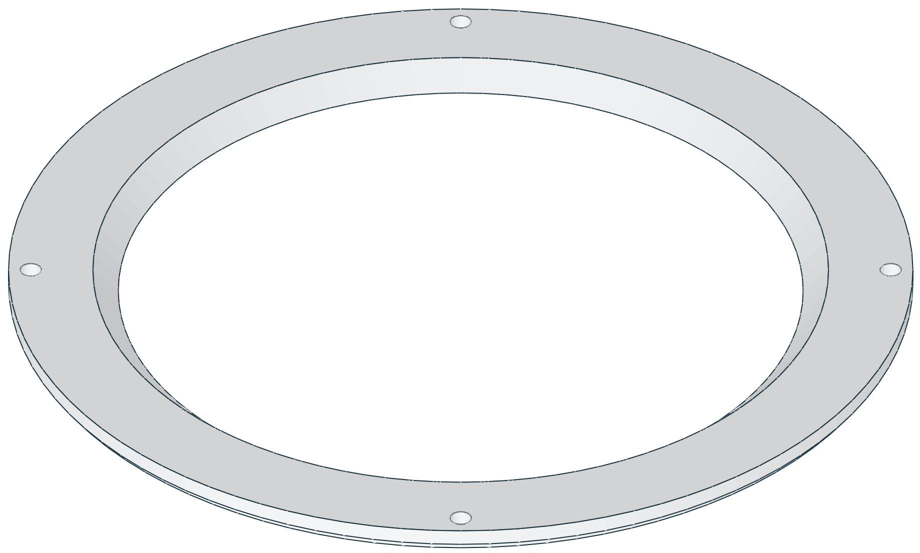
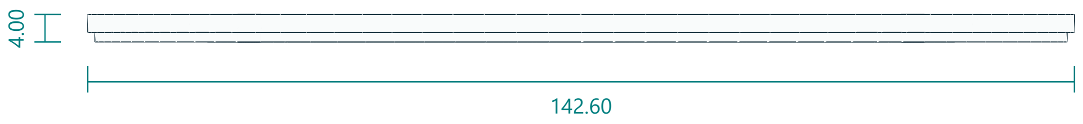


FRONT / XZ



RIGHT / YZ

Interfaces: 142.6 diameter x4; opening108 chamfered to116; screws radius67.8.
Fasteners / retention: 4 M3x12 into shell bosses.
Print orientation: flat front down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes radius67.8 angles45+90n. Rear opening108 chamfered to116 at the front. Rear flange recess145/140.5 by1.4.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.